

Scheduling large robotic cells without buffers

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A *robotic cell* is a manufacturing system that is widely used in industry. A robotic cell contains two or more robot-served machines, repetitively producing a family of similar parts, in a steady state. There are no buffers at or between the machines. Both the robot move cycle and the sequence of parts to produce are chosen in order to minimize the cycle time needed to produce a given set of parts. This objective is also equivalent to throughput rate maximization. In practice, simple robot move cycles that produce one unit are preferred by industry. In an m machine cell for $m \geq 2$, there are $m!$ such cycles that are potentially optimal. Choosing any one of these cycles reduces the cycle time minimization problem to a unique part sequencing problem. We prove the following results in an m machine cell, for any $m \geq 2$. The part sequencing problems associated with these robot move cycles are classified into the following categories: (i) sequence independent; (ii) capable of formulation as a traveling salesman problem (TSP), but polynomially solvable; (iii) capable of formulation as a TSP and unary NP-hard; and (iv) unary NP-hard, but not having TSP structure. As a consequence of this classification, we prove that the part sequencing problems associated with exactly $2m - 2$ of the $m!$ available robot cycles are polynomially solvable. The remaining cycles have associated part sequencing problems which are unary NP-hard.

Keywords: robotic cell, algorithms, computational complexity, traveling salesman problem

1. Introduction

Robots are used in a wide range of applications in manufacturing companies [1, 18]. One important application of robots in manufacturing is their use for material handling in *robotic cells*. A robotic cell is a cluster of production machines arranged within the reach of a robot. The robot unloads parts from, transports them between, and loads them onto, the machines. One of the most general cell layouts is a mobile-

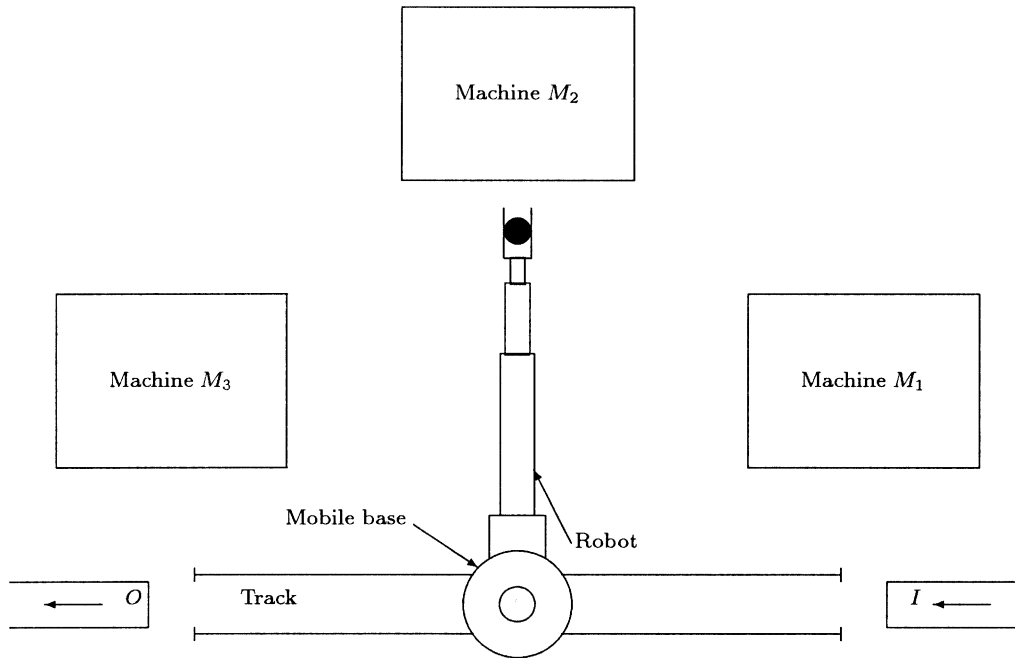


Figure 1. A three-machine mobile-robot cell.

robot cell [11], as illustrated in figure 1. The machines are served by a central robot. The robot travels along a floor-mounted track, and the robot arm rotates and moves linearly, to handle the inter-machine movements of the parts. Each part being processed passes through the same sequence of locations from the input hopper (I), through machines M_1 , M_2 and M_3 , and finally into the output hopper (O), as in a classical flowshop. After loading a part on a machine, the robot either waits for it to finish processing, or moves to another machine to unload another part when that part finishes processing, or moves to the input hopper to pick up a new part.

We consider mobile-robot cells with no buffers at or between the machines. Such cells are common in practice [1, 31]. Since there are no storage buffers, any part in the cell is always either on one of the machines, or being handled by the robot. Neither the machines nor the robot can be in possession of more than one part at any time. Where a variety of parts is being produced, the processing times on a given machine may vary. Parts may have different processing times on different machines. The processing of any part on a machine is nonpreemptive. The robot travel times, and times to pick up, load, unload, and drop parts, are considered to be independent of the part being moved. Rotational and radial movements of the robot may occur simultaneously, but sometimes additional linear movement is necessary after those movements have been completed. Also, travel times are assumed to be independent of direction.

In most applications, robotic cells are used in the repetitive production of a minimal part set (MPS). An MPS is the smallest possible set of parts having the same proportions as the day's production target. For example, if the total day's production is 3,000 units of type A, 1,000 units of type B, and 2,500 units of type C, the MPS consists of 13 parts: six of A, two of B, and five of C. The MPS contains k different part-types which must be produced in given ratios. The objective is the minimization of the steady-state cycle time to produce the minimal part set, or equivalently, the maximization of throughput rate. Two decisions need to be made: finding the robot move cycle to use, and choosing the sequence of parts to produce. If we fix a robot move cycle, then the unique part sequencing problem that remains is usually nontrivial. We say that the part sequencing problem is *associated with* that robot move cycle. The size of an MPS can in practice be larger than 50 parts [38], thus the efficiency of part sequencing procedures becomes very important. Lee and Posner [24] and Hall et al. [15] study the periodic scheduling of a minimal part set in job shop environments, and describe properties of optimal schedules as well as algorithms for cycle time and makespan minimization, and intractability proofs.

Hall [12] provides an introduction to the use of operations research techniques for the design, planning, control and analysis of robotic systems. Reviews of the literature related to the scheduling of robotic cells appear in Sethi et al. [34] and Hall et al. [13]. Crama [4] reviews production planning and scheduling models in automated manufacturing systems that include material handling devices. It is important to understand the relationships between scheduling problems arising in different material handling environments, since this understanding provides insights and also useful results in some cases. For example, the operation of a bidirectional automated guided vehicle (AGV) and that of a robot are similar from a scheduling perspective. Therefore, some of the results in AGV studies are directly applicable to robotic cell scheduling. Ganesharajah et al. [6] review scheduling problems in AGV-served manufacturing systems and relate AGV scheduling problems to the scheduling problems encountered in various material handling environments. It has to be noted that the results obtained for robotic cells are directly applicable to the case of a bidirectional AGV that is used in a line layout. For example, an m machine ($m \geq 2$), single line, bidirectional single AGV problem is mathematically equivalent to the m machine robotic cell problem studied both in this paper, and also by Sethi et al. [34], Logendran and Sriskandarajah [29], and Hall et al. [13, 14].

Sethi et al. [34] prove that there are exactly $m!$ *one-unit robot move cycles* (i.e., robot routing policies) in an m machine cell without buffers producing a single-part type. In such a cycle, the cell returns to the same state after the production of every unit (or part). A specific robot move cycle for $m = 2$, studied in their work, is also considered for the control of a unidirectional AGV system by Kise et al. [22]. For an AGV system which is a two-machine flowshop loop with no buffers for work-in-process, they provide an $O(n^3)$ time algorithm, based on the algorithm of Gilmore and Gomory [9], to find a sequence of n parts that minimizes the makespan. Cheng et al. [3]

consider an m machine automated flowshop problem in which an AGV sends at most one part at a time. They design a heuristic based on an approximation to an $m + 1$ machine classical flowshop problem, and also describe and test a branch and bound algorithm.

Hall et al. [13] follow the standard classification scheme of Graham et al. [10] to describe a scheduling problem. The scheduling environment, job characteristics or requirements, and the optimality criterion, respectively, are represented by ψ_1 , ψ_2 and ψ_3 in the form $\psi_1|\psi_2|\psi_3$. An example of the problems considered in this paper is: $MRC4|no-buffer, k \geq 2, S_i|C_t$, where $MRCm$ represents a mobile-robot cell with m machines and no buffers; k is the number of part types being produced; S_i is a specific robot move cycle over which the problem is being solved; and finally, C_t is the average cycle time for the repetitive manufacture of an MPS.

We now discuss several applications that are related to our models. We begin with a discussion of an in-line robot cell (*IRC*). Phillips and Unger [32], Shapiro and Nuttle [35], Song et al. [36], and Lei and Wang [26] discuss an application of this model to a single hoist electroplating line. Here, m processing tanks (i.e., machines) arranged in series apply chemical or plating treatments on printed circuit boards (i.e., parts or jobs). The hoist (i.e., robot) handles the inter-tank movements of the jobs. A tank can process one job at a time, and a job cannot pause between the tanks, due to deterioration of the job while exposed to the atmosphere. Each job must be immersed in tank M_i for some time p_i , where $u_i \leq p_i \leq v_i$ for given constants u_i and v_i , $i = 1, \dots, m$. We may think of this problem as a general problem, where the processing times of parts on each machine fall within an interval. Hertz et al. [19] discuss a real industrial example where each job consists of 16 operations in a 12-machine jobshop. If $v_i = u_i$ for all i , then we have a no-wait environment [27]. If $v_i = \infty$ for all i , then we have a bufferless environment like the one considered in this paper. In this case, the processing time of part i is always $p_i = u_i$, and after processing a job can remain on the machine for any amount of time before the robot removes it. This is not possible in the no-wait environment, where a job must be processed for time $p_i = u_i$ exactly, and then immediately removed and loaded onto the next machine.

Livshits et al. [28] discuss algorithms and complexity issues for the single hoist problem. Lei and Wang [25] consider a problem where the distance (travel time) between machines is given by an arbitrary distance matrix and the upper bound, v_i , on the processing time is relaxed (i.e., $v_i = \infty$ for all i). They show that the resulting problem, $IRCM|no-buffer, k = 1, 1-unit|C_t$, is unary NP-hard, since it is equivalent to a general traveling salesman problem (see Garey and Johnson [7], for related definitions). For a specific distance matrix similar to the one defined in section 2, Crama and van de Klundert [5] provide a polynomial time dynamic programming algorithm for minimizing cycle time over all one-unit cycles in $RCCm|no-buffer, k = 1, 1-unit|C_t$, where $RCCm$ denotes a robot-centered cell with m machines. They show that this problem is a special case of the traveling salesman problem which can be solved by an $O(m^3)$ time algorithm. The conjecture of Sethi et al. [34] states that the policy of

repeating one-unit cycles dominates the use of multiple-unit cycles in the minimization of cycle time for single-part type problems in bufferless robotic cells ($MRCm|no-buffer, k=1|C_t$). They show that this property holds for $m=2$. This property also holds for $m=3$ when $k=1$ (see [2, 37]). The conjecture remains open for $m \geq 4$ when $k=1$.

Kats and Mikhaletsky [21] study the no-wait robotic cell problem ($RCCm|no-wait, k=1, 1-unit|C_t$) with an arbitrary distance matrix, and show that the number of one-unit cycles in this problem is $O(m^3)$. They also suggest an algorithm for finding the optimal one-unit cycle, but do not estimate its running time. Mikhalevich et al. [30] investigate that algorithm and show its time complexity to be $O(m^5)$. Levner et al. [27] use a network flow formulation to develop an $O(m^3)$ time algorithm for the no-wait problem, $RCCm|no-wait, k=1, 1-unit|C_t$, again with an arbitrary distance matrix. However, problem $RCCm|no-wait, k=1, q-units|C_t$ remains open as to complexity for $q \geq 2$.

The relationship between the bufferless scheduling environment studied here and the no-wait environment (which is also bufferless) requires clarification. No-wait scheduling problems are a more constrained version of bufferless problems. First, consider single-part type problems. As shown by Sethi et al. [34], there are $m!$ available one-unit cycles in a bufferless environment. However, for feasibility in the no-wait problem, a part must not stay on a machine after processing has finished, which in general may eliminate many of these available cycles. Levner et al. [27], as discussed above, identify an optimal no-wait cycle. The no-wait requirement also complicates multiple-part type problems, because meeting that requirement depends on both the robot move cycle and the part sequence. Thus, the number of feasible one-unit cycles is in general even smaller than in the single-part type case. There are apparently no studies on no-wait robotic cell problems with multiple-part types. It follows from this discussion that results do not transfer easily between the bufferless and no-wait environments.

We now briefly describe some recent developments related to the scheduling of multiple parts in bufferless robotic cells, the topic of this paper. In multiple-part type production, we need to produce a set of parts (MPS) repetitively. We consider a *production cycle*, which produces a set of parts repetitively using n repetitions of a one-unit robot move cycle, where n is the number of parts in the set. A one-unit robot move cycle is of particular importance, both in practice and in our study. While much more complicated robot move cycles are available, one-unit cycles are the easiest to understand, implement and control. For these reasons, they are widely used in industry. Furthermore, the study of one-unit cycles provides insight about the behavior of cycles producing two or more units which are made up of portions of one-unit cycles.

Sethi et al. [34] solve the part sequencing problem associated with both the one-unit cycles in a two-machine cell producing multiple-part types. They show that for one cycle the problem is trivial, whereas for the other cycle it is equivalent to a special case of the traveling salesman problem (TSP), which can be solved optimally using the polynomial time algorithm of Gilmore and Gomory [9]. Hall et al. [13] show that

the optimal solution to the multiple-part types problem in a two-machine cell is not generally given by a one-unit cycle. They develop an efficient algorithm that simultaneously optimizes the robot move cycle and part sequencing problems. Hall et al. [13,14] resolve the complexity of the part sequencing problem associated with each of the six available one-unit robot move cycles in a three-machine cell. For four of these cycles, the associated part sequencing problem is solvable in polynomial time, but for the other two cycles it is unary NP-hard. Kamoun et al. [20] develop and test several heuristics for the general three- and four-machine cell, multiple-part type problems. They also identify and test computationally effective heuristics for the general problem of determining both the robot move cycle and the part sequence. Suggestions for extending those approaches to larger cells are also provided.

This paper generalizes the complexity results of Hall et al. [13,14]. In particular, we determine the tractability of the part sequencing problems associated with various robot move cycles in an m machine cell without buffers. We consider only one-unit robot move cycles that are feasible and potentially optimal. We show that, of the $m!$ cycles of this type in an m machine cell, exactly $2m - 2$ have an associated part sequencing problem which is efficiently solvable, whereas the part sequencing problems associated with the other cycles are intractable. Among the intractable part sequencing problems, we make the computationally important distinction between those that can be formulated as a TSP, and those that cannot.

The remainder of this paper is organized as follows. In section 2, we specify our notation for robotic cell scheduling problems. In section 3, we define one-unit cycles and show how to derive their cycle time formulas. In section 4, we classify into four levels of tractability the part sequencing problems associated with one-unit cycles in an m machine mobile-robot cell producing multiple part-types. Finally, section 5 provides a conclusion and some suggestions for future work.

2. Notation

The following notation is used to describe a robotic cell.

$P_1, \dots, P_k$	: the part-types to be produced.
$r_1, \dots, r_k$	: the minimal quantities of parts to be produced.
MPS	: a minimal part set consisting of r_i parts of type P_i , $i = 1, \dots, k$.
$n = r_1 + \dots + r_k$	: the total number of parts to be produced in the MPS.
$M_1, \dots, M_m$	: the machines in the robotic cell, in processing order.
I	: the input hopper.
O	: the output hopper.
σ	: the sequence of parts of an MPS that are processed repetitively.
$P_{\sigma(j)}$	: the j th part in the sequence σ to be produced.

$a_{\sigma(j)}$ (respectively, $b_{\sigma(j)}, c_{\sigma(j)}, d_{\sigma(j)}$)	: the processing time of the j th part in sequence σ on machine M_1 (resp., M_2, M_3, M_4).
$p_{j,i}$	: the processing time of part P_j on machine M_i .
δ_0	: the time taken by additional linear robot movement when traveling between two consecutive machines M_i and M_{i+1} , $1 \leq i \leq m-1$, after rotational movement has been completed.
δ_i	: the time taken by a rotational robot movement when traveling between two consecutive machines M_{i-1} and M_i , $1 \leq i \leq m+1$, where we take I to be machine M_0 , and O to be machine M_{m+1} .
ε_1	: the time taken by the robot to pick up a part at I .
ε_{2i}	: the time taken by the robot to load a part onto machine M_i .
ε_{2i+1}	: the time taken by the robot to unload a part from machine M_i .
ε_{2m+2}	: the time taken by the robot to drop a part at O .
γ	: the time saved by not stopping at an intermediate point during a robot movement between two nonconsecutive machines.
M_i^-	: denotes the following robot activity: <i>load a part on machine M_i</i> .
M_i^+	: denotes the following robot activity: <i>unload a finished part from machine M_i</i> .
E	: $(\chi_1, \dots, \chi_m, M_h^j)$ is the current state of the system, where $\chi_i = \phi$ (respectively, Ω) if machine M_i is free (respectively, occupied by a part), and the robot has just loaded (respectively, unloaded) machine M_h if $j = -$ (respectively, $+$), for $0 \leq h \leq m+1$, where we let $M_0 = I$ and $M_{m+1} = O$.
E_0	: a state of the cell in which all machines except M_i are free, for some $1 < i \leq m$, and the robot has just completed loading a part on M_i , and is about to move to the input hopper, I .
E_{00}	: a state of the cell in which all machines are free, and the robot is about to pick up a part at the input hopper, I .
$T_{j(\sigma)}$	: $T_{j\sigma(1)\sigma(2)}^h + T_{j\sigma(2)\sigma(3)}^h + \dots + T_{j\sigma(n)\sigma(1)}^h$, where $T_{j(\sigma)}$ is the cycle time of an MPS processed in the sequence σ , and $T_{j\sigma(i)\sigma(i+1)}^h$ is the time between the loading of part $P_{\sigma(i)}$ on machine M_h , and the loading of part $P_{\sigma(i+1)}$ on machine M_h , using robot move cycle $S_{j,m}$, where $h \in \{1, \dots, m\}$ and $j \in \{1, \dots, m!\}$. The robot move cycles are defined in section 3.
w_h^i	: the waiting time of the robot before unloading part $P_{\sigma(i)}$ at machine M_h , $h \in \{1, \dots, m\}$.

The robot travel time between locations x and y is denoted by $\ell(x, y)$. For example, in a three-machine mobile-robot cell, $\ell(I, M_1) = \delta_1$, $\ell(M_1, M_2) = \delta_0 + \delta_2$,

$\ell(M_2, M_3) = \delta_0 + \delta_3$, $\ell(M_3, O) = \delta_4$, $\ell(O, I) = 2\delta_0 + \delta_1 + \delta_2 + \delta_3 + \delta_4 - 3\gamma$, $\ell(O, M_1) = 2\delta_0 + \delta_2 + \delta_3 + \delta_4 - 2\gamma$, and $\ell(M_2, I) = \delta_0 + \delta_1 + \delta_2 - \gamma$. The velocity of the robot arm as it moves from one point to another has three different phases: acceleration, constant velocity, and deceleration. For example, the travel time from I to M_2 directly is γ units less than the travel time from I to M_1 plus that from M_1 to M_2 , since the latter requires additional acceleration and deceleration phases. Similarly, the travel time to move directly from I to O is 3γ units less than the travel time from I to M_1 plus those from M_1 to M_2 , M_2 to M_3 , and M_3 to O .

3. One-unit cycles and cycle time derivations

Readers who are familiar with one-unit robot move cycles and how to derive their cycle time formulas may wish to omit this section. Sethi et al. [34] define one-unit cycles in the context of single-part type production, and prove that there are exactly $m!$ potentially optimal one-unit cycles in an m machine cell ($MRCm|no-buffer, k = 1, 1-unit|C_1$). Here, we discuss how to develop these cycles for $m = 2, 3$ and 4 . In such a cell, we define $m + 1$ *basic activities* as follows:

M_i^- : Load a part on M_i , for $i = 1, \dots, m$;

M_m^+ : Unload a finished part from M_m .

We now provide a brief tutorial on how to interpret a robot move cycle composed of the above basic activities. A convenient point from which to begin the interpretation is with the loading of a part on machine M_1 , denoted M_1^- . Consider the sequence, ξ , of basic activities from one occurrence of M_1^- to the next such occurrence. Within ξ , there must be exactly one occurrence of each activity $M_2^-, \dots, M_m^-, M_m^+$. Which machines are occupied by parts at the time when M_1^- occurs can be determined as follows. For $i = 2, \dots, m - 1$, machine M_i is occupied by a part when M_1^- occurs if and only if M_{i+1}^- precedes M_i^- in ξ . Also, machine M_m is occupied by a part when M_1^- occurs if and only if M_m^+ precedes M_m^- in ξ . This interpretation provides a starting point for the cycle. Since exactly which machines are occupied is known, the sequence of basic activities following M_1^- uniquely determines the ordering of loading and unloading of the machines, and thus also the robot movements between activities. Examples for two- and three-machine robotic cells are given below.

Note that, in a one-unit cycle, every basic activity must be carried out exactly once. Moreover, since in an optimal cycle we require that the robot move path is as short as possible, any two consecutive activities uniquely determine the robot moves between those activities. Therefore, a cycle can be uniquely described by a permutation of the above $m + 1$ activities. The following are the available robot move cycles for $m = 2$:

$$S_{1,2}: \{M_2^+, M_1^-, M_2^-, M_2^+\}, \quad S_{2,2}: \{M_2^+, M_2^-, M_1^-, M_2^+\}.$$

We now interpret these two cycles. In cycle $S_{1,2}$, we observe that, following M_1^- in ξ , activity M_2^- precedes M_2^+ . Therefore, we know that machine M_2 is free when M_1^- occurs. By contrast, in cycle $S_{2,2}$, activity M_2^+ precedes M_2^- . Therefore, we know that machine M_2 is occupied by a part when M_1^- occurs.

The six robot moves cycles for $m = 3$ can be developed from the above two cycles, by first replacing the starting and ending activities by M_3^+ , as follows:

- (1) $\{M_3^+, M_1^-, M_2^-, M_3^+\}$,
- (2) $\{M_3^+, M_2^-, M_1^-, M_3^+\}$.

Each of the last two cycles generates three cycles in a three-machine cell, depending upon where the M_3^- activity is inserted, thus creating a total of six cycles, as follows:

$$\begin{aligned} S_{1,3}: \{M_3^+, M_1^-, M_2^-, M_3^-, M_3^+\}, & \quad S_{2,3}: \{M_3^+, M_1^-, M_3^-, M_2^-, M_3^+\}, \\ S_{3,3}: \{M_3^+, M_3^-, M_1^-, M_2^-, M_3^+\}, & \quad S_{4,3}: \{M_3^+, M_2^-, M_3^-, M_1^-, M_3^+\}, \\ S_{5,3}: \{M_3^+, M_2^-, M_1^-, M_3^-, M_3^+\}, & \quad S_{6,3}: \{M_3^+, M_3^-, M_2^-, M_1^-, M_3^+\}. \end{aligned}$$

We briefly interpret two examples of these cycles. In cycle $S_{5,3}$, since M_3^- precedes M_2^- in ξ , machine M_2 must be occupied by a part when M_1^- occurs; also, since M_3^- precedes M_3^+ , machine M_3 must be free at that time. In cycle $S_{6,3}$, since M_3^- precedes M_2^- in ξ , machine M_2 must be occupied by a part when M_1^- occurs; also, since M_3^- precedes M_3^+ , machine M_3 must be occupied by a part at that time.

Each of the above three machine cycles generates four cycles in a four-machine cell, depending upon where the M_4^- activity is inserted, thus creating a total of 24 cycles, as shown in appendix A.

We now describe a method for computing the cycle time of a robot move cycle. In multiple-part type production, we need to produce a set of parts (MPS) repetitively. We consider a production cycle, as defined in section 1.

In order to derive the cycle time expressions for the production cycles, we define the state of the system by $E = (\chi_1, \dots, \chi_m, M_h^j)$, as described in section 2. It should be noted that, for the purpose of deriving cycle times, this state space representation is complete. One might argue that, whenever $\chi_i = \Omega$ for $i \in \{1, 2, \dots, m\}$, the remaining processing time, τ_i , of the part being processed at machine M_i must be included in the state description. Since we limit ourselves to finding an optimal part sequence over a class of production cycles which produce a set of parts repetitively using a fixed robot move cycle, the inclusion of τ_i in the state space is redundant, as discussed by Hall et al. [14]. They show that, in steady-state production of multiple-part types, the cell returns to the same state after producing either one or two MPSs. More specifically, a robot move cycle containing a state in which all the machines are free, except one where the robot has just loaded the part, requires only one MPS to define a production cycle, while others may require two MPSs (see [14] for details). We now derive the cycle times for producing a set of parts (an MPS) under cycles $S_{1,2}$ and $S_{2,2}$ in $MRC2|k \geq 2, 1\text{-unit}|C_t$.

Starting from the initial state $E = (\phi, \Omega, M_2^-)$, where the robot has just loaded part $P_{\sigma(i)}$ on M_2 and M_1 is free, robot move cycle $S_{1,2}$ includes the following activities: wait for $P_{\sigma(i)}$ at M_2 : $(b_{\sigma(i)})$, unload: (ε_5) , move to O : (δ_3) , drop $P_{\sigma(i)}$: (ε_6) , move to I : $(\delta_0 + \delta_1 + \delta_2 + \delta_3 - 2\gamma)$, pick up $P_{\sigma(i+1)}$: (ε_1) , move to M_1 : (δ_1) , load: (ε_2) , wait for $P_{\sigma(i+1)}$: $(a_{\sigma(i+1)})$, unload: (ε_3) , move to M_2 : $(\delta_0 + \delta_2)$, and load: (ε_4) . Thus, $T_{1\sigma(i)\sigma(i+1)}^2 = b_{\sigma(i)} + \varepsilon_5 + \delta_3 + \varepsilon_6 + (\delta_0 + \delta_1 + \delta_2 + \delta_3 - 2\gamma) + \varepsilon_1 + \delta_1 + \varepsilon_2 + a_{\sigma(i+1)} + \varepsilon_3 + (\delta_0 + \delta_2) + \varepsilon_4$. That is,

$$T_{1\sigma(i)\sigma(i+1)}^2 = \rho + b'_{\sigma(i)} + a'_{\sigma(i+1)},$$

where $a'_i = a_i + \varepsilon_1 + \varepsilon_2 + 2\delta_1 - \gamma$, $b'_i = b_i + \varepsilon_5 + \varepsilon_6 + 2\delta_3 - \gamma$, $i = 1, \dots, n$, and $\rho = \varepsilon_3 + \varepsilon_4 + 2\delta_0 + 2\delta_2$.

The time to produce the parts in an MPS in the sequence σ under robot move cycle $S_{1,2}$ is

$$T_{1(\sigma)} = \sum_{i=1}^n T_{1\sigma(i)\sigma(i+1)}^2 = n\rho + \sum_{i=1}^n (a'_i + b'_i).$$

Starting from the initial state $E = (\phi, \Omega, M_2^-)$, where the robot has just loaded part $P_{\sigma(i)}$ on M_2 and M_1 is free, robot move sequence $S_{2,2}$ includes the following activities: move to I : $(\delta_0 + \delta_1 + \delta_2 - \gamma)$, pick up part $P_{\sigma(i+1)}$: (ε_1) , move to M_1 : (δ_1) , load: (ε_2) , move to M_2 : $(\delta_0 + \delta_2)$, if necessary, wait for $P_{\sigma(i)}$: (w_2^i) , unload: (ε_5) , move to O : (δ_3) , drop $P_{\sigma(i)}$: (ε_6) , move to M_1 : $(\delta_0 + \delta_2 + \delta_3 - \gamma)$, if necessary wait for $P_{\sigma(i+1)}$: (w_1^{i+1}) , unload: (ε_3) , move to M_2 : $(\delta_0 + \delta_2)$, and load: (ε_4) . Therefore,

$$\begin{aligned} T_{2\sigma(i)\sigma(i+1)}^2 &= (\delta_0 + \delta_1 + \delta_2 - \gamma) + \varepsilon_1 + \delta_1 + \varepsilon_2 + (\delta_0 + \delta_2) \\ &\quad + w_2^i + \varepsilon_5 + \delta_3 + \varepsilon_6 + (\delta_0 + \delta_2 + \delta_3 - \gamma) \\ &\quad + w_1^{i+1} + \varepsilon_3 + (\delta_0 + \delta_2) + \varepsilon_4, \end{aligned}$$

where

$$w_1^{i+1} = \max\{0, a_{\sigma(i+1)} - w_2^i - \varepsilon_5 - \varepsilon_6 - 2\delta_0 - 2\delta_2 - 2\delta_3 + \gamma\}$$

and

$$w_2^i = \max\{0, b_{\sigma(i)} - \varepsilon_1 - \varepsilon_2 - 2\delta_0 - 2\delta_1 - 2\delta_2 + \gamma\}.$$

By combining the expressions for w_1^{i+1} and w_2^i , we obtain the following result:

$$T_{2\sigma(i)\sigma(i+1)}^2 = \rho + \max\{\mu, b'_{\sigma(i)}, a'_{\sigma(i+1)}\},$$

where $\mu = 2\sum_{i=0}^3 \delta_i + \varepsilon_1 + \varepsilon_2 + \varepsilon_5 + \varepsilon_6 - 2\gamma$.

The time to produce the parts in an MPS in the sequence σ under robot move cycle $S_{2,2}$ is

$$T_{2(\sigma)} = \sum_{i=1}^n T_{2\sigma(i)\sigma(i+1)}^2 = n\rho + \sum_{i=1}^n \max\{\mu, b'_{\sigma(i)}, a'_{\sigma(i+1)}\}.$$

The cycle time derivations for cycles $S_{1,3}, \dots, S_{6,3}$ in $MRC3|k \geq 2, 1\text{-unit}|C_t$ are shown in appendix B.

4. Part sequencing problems in an m machine cell

In this section, we consider problems where an m machine robotic cell is producing multiple-part types, using a one-unit cycle, or $MRCm|no-buffer, k \geq 2, 1-unit|C_t$, where $m \geq 2$. Our purpose is to characterize which of the available cycles will permit the efficient identification of an optimal part sequence.

We need to point out that the robot may perform two kinds of waiting in a cell. We refer to these as Type 1 and Type 2 waiting. In Type 1 waiting, the robot loads a part on a machine and waits there throughout processing; therefore, the waiting time is exactly the processing time of that part on that machine. In Type 2 waiting, the robot loads a part and performs other activities during its processing. The robot later returns to unload the part, sometimes needing to wait until processing finishes. Here, the waiting time is only from the time when the robot returns until the completion of processing. Depending upon the data, a Type 2 waiting time may be zero. Note that all the w_h^i terms used in the cycle time derivations in section 3 are Type 2 waiting times before the robot unloads part $P_{\sigma(i)}$ at machine M_h , where $h \in \{1, \dots, m\}$. Not all robot move cycles have the same number of Type 2 waiting times. For example, a cycle in which the robot stays with a part during processing at each machine will have no Type 2 waiting times, but all other robot move cycles will have at least one Type 2 waiting time. Formally, a given robot move cycle has a Type 2 waiting time at machine M_i if and only if the robot performs some activities between the loading of a part on M_i and the unloading of that part from M_i in that cycle. We let $\mathcal{N}^i$ (respectively, $\mathcal{N}_j$) denote the number of Type 2 waiting times on machines $M_1, \dots, M_i$ (resp., $M_j, \dots, M_m$) in a given robot move cycle, where $i \leq m$ and $j \geq 1$. We also let $\mathcal{N} = \mathcal{N}^m = \mathcal{N}_1$ denote the total number of Type 2 waiting times in a robot move cycle. In the remainder of the paper, when we refer to waiting times, these are of Type 2.

Usually, the larger is the number of waiting times in a robot move cycle, the less tractable is the associated part sequencing problem. This is because the waiting times are usually interdependent, and thus processing times of more than one part contribute to one waiting time. This recursive structure of the waiting times makes the part sequencing problem more difficult to solve. However, exceptions may occur where waiting times have a recursive structure but depend on only one other part or waiting time. We classify the one-unit robot move cycles in an m machine cell, for $m \geq 2$, according to the tractability of their associated part sequencing problems, as follows:

Class U : sequence independent (trivially solvable).

Class V : TSP models.

Class W : unary NP-hard, but not having TSP structure.

Class V can be further divided into two subclasses:

Class $V1$: TSP special cases (solvable in polynomial time using the algorithm of Gilmore and Gomory [9]).

Class $V2$: unary NP-hard TSP models.

The following subsections examine these classes in turn. Before considering them, we define several easily verifiable conditions, C1–C3, on a robot move cycle. As we will prove, these conditions determine the complexity of the part sequencing problem associated with that cycle.

C1 : $\mathcal{N} = 0$.

C2a : $\mathcal{N} = 2$.

C2b : at least one Type 2 waiting time occurs at machine M_1 or M_m .

C3 : the cycle reaches state E_0 at machine M_q , for some $1 < q \leq m$.

The following discussion is needed to clarify the relationships between Classes U , $V1$, $V2$ and W , and conditions C1–C3. Clearly, condition C1 is mutually incompatible with either condition C2a or condition C2b. Since the definition of E_0 implies that $\mathcal{N} \geq 1$, condition C1 is mutually incompatible with condition C3. Finally, as will be shown in corollary 1, condition C2a implies condition C3. These relationships are illustrated in table 1, where infeasible combinations of conditions are omitted. Note

Table 1

Classes of robot move cycles.

Class	U	$V1$	$V2$	$V2$	$V2$	W	W
C1	Yes	No	No	No	No	No	No
C2a	No	Yes	Yes	No	No	No	No
C2b	No	Yes	No	Yes	No	Yes	No
C3	No	Yes	Yes	Yes	Yes	No	No

that the three (respectively, two) columns for Class $V2$ (resp., W) represent the different conditions under which this class of cycles occurs. As will be discussed in the proof of lemma 12, there are no robot move cycles in which $\mathcal{N} = 1$.

It should be noted that cycles $S_{1,2}$ and $S_{1,3}$ are U -cycles, in which $\mathcal{N} = 0$. Cycles $S_{2,2}$, $S_{3,3}$, $S_{4,3}$ and $S_{5,3}$ are $V1$ -cycles, since they contain an E_0 state, $\mathcal{N} = 2$, and one of the Type 2 robot waiting times occurs either on the first machine or on the last machine. However, cycles $S_{2,3}$ and $S_{6,3}$ are W -cycles, since they do not contain an E_0 state and $\mathcal{N} > 2$.

4.1. Class U : Sequence independent problems

We begin with a preliminary result.

Lemma 1. In any one-unit robot move cycle that satisfies condition C1 in an m machine cell, at most one part is in the cell at any point in time.

Proof. If $\mathcal{N} = 0$, then when the robot is about to pick up a part $P_{\sigma(j)}$ at I , the cell must be in state E_{00} , where all machines are free. Since the cycle repeats after one unit, there must also be a point in time at which all machines are free and the robot is about to pick up part $P_{\sigma(j+1)}$ at I . Thus, part $P_{\sigma(j)}$ must leave the cell before part $P_{\sigma(j+1)}$ enters. $\square$

This leads to the following result.

Lemma 2. The optimal part sequence in an m machine cell can be found trivially in one robot move cycle that satisfies condition C1.

Proof. It follows from lemma 1 that, in a robot move cycle with $\mathcal{N} = 0$, the cycle time can be written as the sum of the processing times of the parts, plus the sum of movement times of the robot. Since both are independent of the part sequence, all part sequences provide the same cycle time. It also follows from lemma 1 that there is exactly one U -cycle in an m machine cell, since the robot waits at every machine while a part is being processed, which defines a unique robot move cycle. $\square$

4.2. Class V1: Special cases of the TSP

Here we show that, if a robot move cycle satisfies conditions C2a and C2b, then the associated part sequencing problem can be formulated as a special case of the TSP and solved using the algorithm of Gilmore and Gomory [9]. The robot move cycles that satisfy these conditions are called V1-cycles.

Lemma 3. The optimal part sequence in an m machine cell can be found in $O(n \log n)$ time in $2m - 3$ robot move cycles that satisfy conditions C2a and C2b.

Proof. The robot move cycles satisfying the lemma conditions have waiting times at the following machines: $(M_1, M_2), \dots, (M_1, M_m), (M_2, M_m), \dots, (M_{m-1}, M_m)$. Thus, there are $2m - 3$ such robot move cycles.

We consider two cases. In each case, we show that the cycle time for robot move cycle $S_{j,m}$ can be written as $T_{j(\sigma)} = \sum_{i=1}^n T_{j\sigma(i)\sigma(i+1)}^h$, where $\sum_{i=1}^n T_{j\sigma(i)\sigma(i+1)}^h = Z + \sum_{i=1}^n \max\{g_{\sigma(i+1)}, h_{\sigma(i)}\}$ for some constant Z , and where we let $\sigma(n+1) = \sigma(1)$. The two parameters $g_{\sigma(i)}$ and $h_{\sigma(i)}$ can be associated with city i in the TSP, where the distance between cities i and $i+1$ is $\max\{g_{\sigma(i+1)}, h_{\sigma(i)}\}$. An instance of the TSP with this distance matrix can be solved by the algorithm of Gilmore and Gomory[9].

Case 1: There is a waiting time at machines M_1 and M_s , for some $2 \leq s \leq m$.

If there is not an E_0 state at machine M_s , then the robot must move to M_1 after loading M_s . In this case, since $\mathcal{N} = 2$, two parts will eventually be on M_s simultaneously, a contradiction. Thus, the cycle reaches an E_0 state at M_s . Starting from that state with

part $P_{\sigma(i)}$ at M_s , cycle S_h includes the following activities: move to I: $((s-1)\delta_0 + \sum_{q=1}^s \delta_q - (s-1)\gamma)$, pick up a new part $P_{\sigma(i+1)}$: (ε_1) , move to M_1 : (δ_1) , load $P_{\sigma(i+1)}$ on M_1 : (ε_2) , move from M_1 to M_s : $((s-1)\delta_0 + \sum_{q=2}^s \delta_q - (s-2)\gamma)$, if necessary wait at M_s : (w_s^i) , unload $P_{\sigma(i)}$ from M_s : (ε_{2s+1}) , move to M_{s+1} : $(\delta_0 + \delta_{s+1})$, load $P_{\sigma(i)}$ on M_{s+1} : (ε_{2s+2}) , wait at M_{s+1} : $(p_{\sigma(i),s+1})$, unload: $(\varepsilon_{2s+3}), \dots$, drop $P_{\sigma(i)}$ at O : (ε_{2m+2}) , move to M_1 : $((m-1)\delta_0 + \sum_{q=2}^{m+1} \delta_q - (m-1)\gamma)$, if necessary wait at M_1 : (w_1^{i+1}) , unload $P_{\sigma(i+1)}$ from M_1 : (ε_3) , move to M_2 : $(\delta_0 + \delta_2)$, load $P_{\sigma(i+1)}$ on M_2 : (ε_4) , wait at M_2 : $(p_{\sigma(i+1),2}), \dots$, and load $P_{\sigma(i+1)}$ on M_s : (ε_{2s}) . Thus,

$$T_{h\sigma(i)\sigma(i+1)}^s = \sum_{q=1}^{2m+2} \varepsilon_q + 2 \sum_{q=1}^{m+1} \delta_q + 2 \sum_{q=2}^s \delta_q + 2(m+s-2)\delta_0 - m\gamma - 2s\gamma + 4\gamma \\ + \sum_{t=2}^{s-1} p_{\sigma(i+1),t} + \sum_{t=s+1}^m p_{\sigma(i),t} + w_1^{i+1} + w_s^i,$$

where

$$w_1^{i+1} = \max \left\{ 0, p_{\sigma(i+1),1} - 2(m-1)\delta_0 - 2 \sum_{q=2}^{m+1} \delta_q + (m+s-3)\gamma \right. \\ \left. - \sum_{q=2s+1}^{2m+2} \varepsilon_q - \sum_{t=s+1}^m p_{\sigma(i),t} - w_s^i \right\}$$

and

$$w_s^i = \max \left\{ 0, p_{\sigma(i),s} - 2(s-1)\delta_0 - 2 \sum_{q=1}^s \delta_q + (2s-3)\gamma - \varepsilon_1 - \varepsilon_2 \right\}.$$

Substituting for w_1^{i+1} , we obtain

$$T_{h\sigma(i)\sigma(i+1)}^s = \sum_{q=1}^{2s} \varepsilon_q + 2 \sum_{q=1}^s \delta_q + 2(s-1)\delta_0 - s\gamma + \gamma + \sum_{t=2}^{s-1} p_{\sigma(i+1),t} \\ + \max \left\{ 2(m-1)\delta_0 + 2 \sum_{q=2}^{m+1} \delta_q - (m+s-3)\gamma + \sum_{q=2s+1}^{2m+2} \varepsilon_q + \sum_{t=s+1}^m p_{\sigma(i),t} + w_s^i, p_{\sigma(i+1),1} \right\}.$$

Letting $g_{\sigma(i+1)} = p_{\sigma(i+1),1}$ and

$$h_{\sigma(i)} = 2(m-1)\delta_0 + 2 \sum_{q=2}^{m+1} \delta_q - (m+s-3)\gamma + \sum_{q=2s+1}^{2m+2} \varepsilon_q + \sum_{t=s+1}^m p_{\sigma(i),t} + w_s^i,$$

we obtain the desired form.

Case 2: There is a waiting time at machines M_s , for some $1 \leq s \leq m-1$, and M_m .

The proof that the cycle reaches an E_0 state at machine M_m is similar to that for M_s in case 1. Starting from that state with part $P_{\sigma(i)}$ at M_m , cycle S_j includes the following

activities: move to I: $((m-1)\delta_0 + \sum_{q=1}^m \delta_q - (m-1)\gamma)$, pick up a new part $P_{\sigma(i+1)}: (\varepsilon_1)$, move to $M_1: (\delta_1)$, load $P_{\sigma(i+1)}$ on $M_1: (\varepsilon_2)$, wait at $M_1: (p_{\sigma(i+1),1}), \dots$, load $P_{\sigma(i+1)}$ on $M_s: (\varepsilon_{2s})$, move from M_s to $M_m: ((m-s)\delta_0 + \sum_{q=s+1}^m \delta_q - (m-s-1)\gamma)$, if necessary wait at $M_m: (w_m^i)$, unload: (ε_{2m+1}) , move to O: (δ_{m+1}) , drop $P_{\sigma(i)}$ at O: (ε_{2m+2}) , move to $M_s: ((m-s)\delta_0 + \sum_{q=s+1}^{m+1} \delta_q - (m-s)\gamma)$, if necessary wait at $M_s: (w_s^{i+1})$, unload: (ε_{2s+1}) , move to $M_{s+1}: (\delta_0 + \delta_{s+1})$, load $P_{\sigma(i+1)}$ on $M_{s+1}: (\varepsilon_{2s+2})$, wait at $M_{s+1}: (p_{\sigma(i+1),s+1}), \dots$, and load $P_{\sigma(i+1)}$ on $M_m: (\varepsilon_{2m})$. Thus,

$$T_{j\sigma(i)\sigma(i+1)}^m = \sum_{q=1}^{2m+2} \varepsilon_q + 2 \sum_{q=1}^{m+1} \delta_q + 2 \sum_{q=s+1}^m \delta_q + 2(2m-s-1)\delta_0 + (-3m+2s+2)\gamma \\ + w_m^i + w_s^{i+1} + \sum_{t=1}^{s-1} p_{\sigma(i+1),t} + \sum_{t=s+1}^{m-1} p_{\sigma(i+1),t},$$

where

$$w_m^i = \max \left\{ 0, p_{\sigma(i),m} - \sum_{q=1}^{2s} \varepsilon_q - 2 \sum_{q=1}^m \delta_q - 2(m-1)\delta_0 - \sum_{t=1}^{s-1} p_{\sigma(i+1),t} + (2m-s-2)\gamma \right\}$$

and

$$w_s^{i+1} = \max \left\{ 0, p_{\sigma(i+1),s} - 2 \sum_{q=s+1}^{m+1} \delta_q - 2(m-s)\delta_0 - \varepsilon_{2m+1} - \varepsilon_{2m+2} - w_m^i + (2m-2s-1)\gamma \right\}.$$

Substituting for w_s^{i+1} , we obtain

$$T_{j\sigma(i)\sigma(i+1)}^m = \sum_{q=1}^{2m} \varepsilon_q + 2 \sum_{q=1}^m \delta_q + (2m-2)\delta_0 - m\gamma + \gamma + \sum_{t=1}^{s-1} p_{\sigma(i+1),t} + \sum_{t=s+1}^{m-1} p_{\sigma(i+1),t} \\ + \max \left\{ 2 \sum_{q=s+1}^{m+1} \delta_q + 2(m-s)\delta_0 + \varepsilon_{2m+1} + \varepsilon_{2m+2} - (2m-2s-1)\gamma + w_m^i, p_{\sigma(i+1),s} \right\}.$$

Substituting for w_m^i , we obtain

$$T_{j\sigma(i)\sigma(i+1)}^m = \sum_{q=1}^{2m} \varepsilon_q + 2 \sum_{q=1}^m \delta_q + (2m-2)\delta_0 - m\gamma + \gamma + \sum_{t=s+1}^{m-1} p_{\sigma(i+1),t} \\ + \max \left\{ 2 \sum_{q=s+1}^{m+1} \delta_q + 2(m-s)\delta_0 + \varepsilon_{2m+1} + \varepsilon_{2m+2} - (2m-2s-1)\gamma \right. \\ \left. + \sum_{t=1}^{s-1} p_{\sigma(i+1),t}, p_{\sigma(i+1),s} + \sum_{t=1}^{s-1} p_{\sigma(i+1),t} \right\},$$

$$p_{\sigma(i),m} - \left(\sum_{q=1}^{2s} \varepsilon_q + 2 \sum_{q=1}^m \delta_q + 2(m-1)\delta_0 - (2m-s-2)\gamma \right) \\ + \left(2 \sum_{q=s+1}^{m+1} \delta_q + 2(m-s)\delta_0 + \varepsilon_{2m+1} + \varepsilon_{2m+2} - (2m-2s-1)\gamma \right) \Bigg\}.$$

Letting

$$g_{\sigma(i+1)} = \max \left\{ 2 \sum_{q=s+1}^{m+1} \delta_q + 2(m-s)\delta_0 + \varepsilon_{2m+1} + \varepsilon_{2m+2} - (2m-2s-1)\gamma \right. \\ \left. + \sum_{t=1}^{s-1} p_{\sigma(i+1),t}, p_{\sigma(i+1),s} + \sum_{t=1}^{s-1} p_{\sigma(i+1),t} \right\}$$

and

$$h_{\sigma(i)} = p_{\sigma(i),m} - \sum_{q=1}^{2s} \varepsilon_q + \varepsilon_{2m+1} + \varepsilon_{2m+2} - 2 \sum_{q=1}^m \delta_q + 2 \sum_{q=s+1}^{m+1} \delta_q + 2(1-s)\delta_0 + (s-1)\gamma,$$

we obtain the desired form.

Finally, we note that the algorithm of Gilmore and Gomory [9] can be implemented in $O(n \log n)$ time [23]. $\square$

4.3. Class V2: NP-hard TSP problems

The robot move cycles belonging to this class are called V2-cycles. These cycles satisfy condition C3, but not both conditions C2a and C2b. We will show that the part sequencing problems associated with these robot move cycles are unary NP-hard. However, each of these problems can be formulated as a TSP, which provides significant computational advantages as a result of the numerous optimal and heuristic approaches available for that problem [23]. For example, the heuristic developed by Gendreau et al. [8] can routinely solve a TSP of up to 300 cities (equivalently here, parts) to within 5% of optimality.

Before presenting the following result, we need a definition. It follows from the theory of computational complexity [7] that, if the TSP can be solved in polynomial time, then so also can an NP-complete part sequencing problem of the type considered here. However, such a solution may require a chain of elaborate formulations and transformations that might be hard to implement. Our definition that follows is both more restrictive and more practically useful. We say that a part sequencing problem *can be formulated as a TSP* if and only if the contribution to cycle time from processing part $P_{\sigma(i)}$ immediately before part $P_{\sigma(i+1)}$ is independent of the data for all other jobs

Lemma 4. For any robot move cycle that does not satisfy condition C1 in an m machine cell, the associated part sequencing problem can be formulated as a TSP if and only if the robot move cycle reaches state E_0 at machine M_q , for some $1 \leq q \leq m$.

Proof. We consider each part of the lemma in turn.

($\Rightarrow$) Assume that the robot move cycle reaches state E_0 at machine M_q . Let Z_i (respectively, Z_{i+1}) denote the time at which the robot loads part $P_{\sigma(i)}$ (resp., part $P_{\sigma(i+1)}$) on M_q . Thus, part $P_{\sigma(i-1)}$ must have left machine M_m by time Z_i , and part $P_{\sigma(i+2)}$ must not have reached machine M_1 by time Z_{i+1} . Consequently, $Z_{i+1} - Z_i$ is independent of the processing times of parts other than $P_{\sigma(i)}$ and $P_{\sigma(i+1)}$. It follows that $Z_{i+1} - Z_i$ can be interpreted as the distance from city i to city $i + 1$ in a formulation of the TSP.

($\Leftarrow$) Consider a robot move cycle that does not satisfy condition C1. Since state E_0 is never reached, when part $P_{\sigma(i)}$ is loaded on any machine, either part $P_{\sigma(i-1)}$ or part $P_{\sigma(i+1)}$ is on machine M_j , for some $1 \leq j \leq m$. In the first case, the remaining processing times in the cell at time Z_{i+1} are not, in general, the same as the remaining processing times in the cell at time Z_i , because they depend on the processing times of parts $P_{\sigma(i-1)}$, $P_{\sigma(i)}$ and $P_{\sigma(i+1)}$. In the second case, a similar argument applies for parts $P_{\sigma(i)}$, $P_{\sigma(i+1)}$ and $P_{\sigma(i+2)}$. In either case, we cannot write $Z_{i+1} - Z_i$ as a function of the processing times of parts $P_{\sigma(i)}$ and $P_{\sigma(i+1)}$ only, as required by our definition of a TSP formulation. $\square$

We have shown in lemma 4 that the part sequencing problem associated with any cycle satisfying condition C3 can be formulated as a TSP. We will compute the number of such cycles in an m machine cell. In order to do so, we need the preliminary results given in the two lemmas which follow.

Lemma 5. In an m machine cell, there exists a robot move cycle that does not satisfy condition C1 and that reaches state E_0 with the robot at machine M_q , for some $1 \leq q \leq m$, if and only if there are $\mathcal{N}^{q-1}$ waiting times on machines $M_1, \dots, M_{q-1}$, one on machine M_q , and $\mathcal{N}^{q-1} - 1$ on machines $M_{q+1}, \dots, M_m$, where $\mathcal{N}^{q-1} \in \{1, \dots, \lfloor m/2 \rfloor\}$.

Proof. Consider an arbitrary robot move cycle, S , that reaches state E_0 at machine M_q and that has $\mathcal{N} \geq 1$. By definition of E_0 , there is a waiting time at M_q . Also by definition of E_0 , the robotic cell reaches state E_0 with the robot at machine M_q in a cycle if and only if the following five events occur in the order shown:

- E1: the robot loads part $P_{\sigma(i)}$ on M_q .
- E2: the robot loads part $P_{\sigma(i+1)}$ on M_1 .
- E3: the robot drops part $P_{\sigma(i)}$ at O .
- E4: the robot loads part $P_{\sigma(i+1)}$ on M_q .
- E5: the robot loads part $P_{\sigma(i+2)}$ on M_1 .

We consider each part of the lemma in turn.

($\Leftarrow$) We consider a robot move cycle that does not satisfy condition C1. If parts $P_{\sigma(i)}$ and $P_{\sigma(i+2)}$ are on two of machines $M_1, \dots, M_m$ at any time during this cycle, then event

E5 will precede event E3. Therefore, a necessary condition to reach state E_0 at M_q is that there is never more than one other part on machines $M_1, \dots, M_m$ simultaneously with part $P_{\sigma(i+1)}$.

We consider two cases.

Case 1: $\mathcal{N}_{q+1} \geq \mathcal{N}^{q-1}$. Consider the cell immediately after event E1. From Sethi et al. [34], as discussed at the beginning of section 3, there are m basic activities, which are loadings on machines $M_1, \dots, M_{q-1}, M_{q+1}, \dots, M_m$ and an unloading from machine M_m , between event E1 and event E4. Since event E3 must precede event E4, part $P_{\sigma(i)}$ must be advanced at least $m - q + 1$ times during those m basic activities. Since only parts $P_{\sigma(i)}$ and $P_{\sigma(i+1)}$ are on machines $M_1, \dots, M_m$, $P_{\sigma(i)}$ is advanced $\mathcal{N}^{q-1}$ times from the machines among $M_1, \dots, M_{q-1}$ that have waiting times, not from machine M_q having a waiting time, and $m - q - \mathcal{N}_{q+1}$ times from the machines among $M_{q+1}, \dots, M_m$ that do not have waiting times. Thus, $\mathcal{N}^{q-1} + (m - q - \mathcal{N}_{q+1}) \geq m - q + 1$, a contradiction.

Case 2: $\mathcal{N}_{q+1} \leq \mathcal{N}^{q-1} - 2$. Since it is necessary for event E4 to precede event E5, $P_{\sigma(i+1)}$ must be advanced at least $q - 1$ times during the m basic activities between event E2 and event E5. Since only parts $P_{\sigma(i)}$ and $P_{\sigma(i+1)}$ are on machines $M_1, \dots, M_m$, $P_{\sigma(i+1)}$ is advanced $q - \mathcal{N}^{q-1} - 1$ times from the machines among $M_1, \dots, M_{q-1}$ that do not have waiting times, once from machine M_q having a waiting time, and $\mathcal{N}_{q+1}$ times from the machines among $M_{q+1}, \dots, M_m$ that have waiting times. Thus, $(q - \mathcal{N}^{q-1} - 1) + 1 + \mathcal{N}_{q+1} \geq q - 1$, a contradiction.

It therefore follows that $\mathcal{N}_{q+1} = \mathcal{N}^{q-1} - 1$, and the total number of waiting times is $1 + \mathcal{N}^{q-1} + \mathcal{N}_{q+1} = 2\mathcal{N}^{q-1}$. Thus, $\mathcal{N}$ is even. Since there is a waiting time at M_q , it follows that $\mathcal{N}^{q-1} \geq 1$. Finally, since the number of waiting times cannot exceed the number of machines, it follows that $\mathcal{N}^{q-1} \leq \lfloor m/2 \rfloor$.

($\Rightarrow$) It follows from the above discussion that, if $\mathcal{N}_{q+1} = \mathcal{N}^{q-1} - 1$, then events E1–E5 will occur in the specified order. Therefore, the robotic cell will reach state E_0 with the robot at machine M_q . $\square$

Corollary 1. Condition C2a implies condition C3.

Proof. It follows immediately from the ($\Rightarrow$) part of lemma 5 that, if a robot move cycle satisfies condition C2a with waiting times at machines M_s and M_q , where $1 \leq s < q \leq m$, then that cycle reaches state E_0 at machine M_q . $\square$

It is interesting to note that the converse of corollary 1 is not true (see table 1). For example, robot move cycles $S_{9,4}$ and $S_{10,4}$ have an E_0 state and thus satisfy condition C3, but have $\mathcal{N} = 4$ which does not satisfy condition C2a.

Lemma 6. For a given selection of $\mathcal{N}$ waiting times, with $\mathcal{N} \geq 2$ and even, in an m machine cell, there is exactly one robot move cycle which reaches state E_0 at machine M_q , for some $1 < q \leq m$.

Proof. First, note that the lemma condition implies that the conditions in lemma 5 are satisfied at machine M_q , for some $1 < q \leq m$. Therefore, it follows from the ($\Rightarrow$) part of lemma 5 that there is at least one robot move cycle satisfying the lemma condition.

As discussed at the beginning of section 3, a robot move cycle in an m machine cell is uniquely characterized by its sequence of $m + 1$ basic activities. Therefore, it is sufficient to show that a given selection of $\mathcal{N}$ waiting times among the m machines uniquely determines a sequence of these basic activities. Assume that there are two distinct robot move cycles, S and S' , which both reach state E_0 with the robot having just completed loading part $P_{\sigma(i)}$ on machine M_q , for the same q , $1 < q \leq m$, and which have waiting times on the same subset of machines. We now prove that S and S' are identical, by induction on the number of robot moves after E_0 .

By definition of E_0 , the robot next picks up part $P_{\sigma(i+1)}$ and loads it on M_1 in both cycles S and S' . For the basis of the induction proof, if M_1 has a waiting time in S and S' , then the second robot activity after E_0 in both cycles is to load part $P_{\sigma(i)}$ on machine M_{q+1} , where we let $m + 1 = O$. Otherwise, the second robot activity after E_0 in both cycles is to load part $P_{\sigma(i+1)}$ on machine M_2 . Thus, the states of the cell are identical in S and S' after two activities. For the induction hypothesis, we assume that the states of the cell are identical in S and S' after $2 \leq t \leq m$ robot activities following E_0 . During activity $t + 1$, when the robot loads a part, ($P_{\sigma(i)}$ or $P_{\sigma(i+1)}$), it either waits with it during processing (in both S and S'), or unloads a part from another machine. It follows from the discussion in the ($\Leftarrow$) part of lemma 5 that there is at most one other part ($P_{\sigma(i+1)}$ or $P_{\sigma(i)}$) on any machine. Moreover, from the induction hypothesis, that part is on the same machine in S and S' . Thus, activity $t + 1$ is identical in S and S' for $t + 1 \leq m + 1$. When $t + 1 = m + 1$, the cycle reaches the same E_0 state in S and S' , and the above arguments can be repeated for subsequent parts. It follows that S and S' are identical. $\square$

We have shown in lemma 5 that one-unit robot move cycles which reach state E_0 with the robot at M_q have a specific number of waiting times before and after M_q in the cell. From lemma 6, any selection of these waiting times among the m machines defines exactly one robot move cycle which satisfies condition C3. The following result shows how many such cycles there are.

Lemma 7. In an m machine cell, the number of robot move cycles that do not satisfy condition C1, and in which the associated part sequencing problem can be formulated as a TSP, is

$$\sum_{t=1}^{\lfloor m/2 \rfloor} \binom{m}{2t}.$$

Proof. It follows from lemma 4 and the proof of lemma 5 that, in any robot move cycle which does not satisfy condition C1 and which has an associated part sequencing problem that can be formulated as a TSP, $\mathcal{N} \geq 2$ and even. Thus, $\mathcal{N} = 2t$, where

$t \in \{1, \dots, \lfloor m/2 \rfloor\}$. It follows from lemma 6 that, for each possible value of t , the number of robot move cycles is

$$\binom{m}{2t},$$

hence the result. $\square$

It is important to note that the $2m - 3$ V1-cycles discussed in lemma 3 are included in this number.

We now consider the complexity of part sequencing problems associated with V2-cycles. Hall et al. [13] show that the part sequencing problem associated with cycle $S_{1,3}$ is sequence independent. Further, they show that the part sequencing problems associated with cycles $S_{3,3}$, $S_{4,3}$ and $S_{5,3}$ are solvable in polynomial time. These cycles satisfy conditions C2a and C2b, and they therefore belong to class V1. The part sequencing problems associated with cycles $S_{2,3}$ and $S_{6,3}$ do not satisfy condition C3, and they therefore belong to class W. Thus, there are no V2-cycles in a three-machine cell. For this reason, it is necessary to study four-machine cells, in order to identify a pattern for larger cells.

In a four-machine cell, there are seven robot move cycles which belong to class V. These are cycles $S_{4,4}$, $S_{9,4}$, $S_{10,4}$, $S_{11,4}$, $S_{13,4}$, $S_{14,4}$ and $S_{17,4}$, as described in appendix A. Note that, since they satisfy conditions C2a and C2b, cycles $S_{4,4}$, $S_{11,4}$, $S_{13,4}$, $S_{14,4}$ and $S_{17,4}$ are V1-cycles. We show in this section that the part sequencing problems associated with the V2-cycles $S_{9,4}$ and $S_{10,4}$ are unary NP-hard. All of the NP-hardness results in this paper are proved for the special case $\delta_i = \delta$ for $i = 0, \dots, m + 1$, $\varepsilon_i = \varepsilon$ for $i = 1, \dots, 2m + 2$, and $\gamma = 0$.

The cycle time for the production of n parts in sequence σ under cycle $S_{9,4}$ is derived as follows:

$$T_{9\sigma(i)\sigma(i+1)}^3 = 12\delta + 10\varepsilon + a_{\sigma(i+1)} + d_{\sigma(i)} + w_2^{i+1} + w_3^i,$$

where

$$w_2^{i+1} = \max\{0, b_{\sigma(i+1)} - d_{\sigma(i)} - w_3^i - 6\delta - 4\varepsilon\}$$

and

$$w_3^i = \max\{0, c_{\sigma(i)} - a_{\sigma(i+1)} - 6\delta - 4\varepsilon\}.$$

Since

$$w_2^{i+1} = \max\{0, b_{\sigma(i+1)} - d_{\sigma(i)} - w_3^i - 6\delta - 4\varepsilon\},$$

$$w_2^{i+1} + w_3^i = \max\{0, c_{\sigma(i)} - a_{\sigma(i+1)} - 6\delta - 4\varepsilon, b_{\sigma(i+1)} - d_{\sigma(i)} - 6\delta - 4\varepsilon\}.$$

Therefore, we can write

$$T_{9(\sigma)} = \sum_{i=1}^n T_{9\sigma(i)\sigma(i+1)}^3,$$

where

$$T_{9\sigma(i)\sigma(i+1)}^3 = \max\{a'_{\sigma(i+1)} + d'_{\sigma(i)}, c'_{\sigma(i)} + d'_{\sigma(i)}, a'_{\sigma(i+1)} + b'_{\sigma(i+1)}\},$$

$$a'_i = a_i + 6\delta + 5\varepsilon, b'_i = b_i + \varepsilon, c'_i = c_i + \varepsilon \quad \text{and} \quad d'_i = d_i + 6\delta + 5\varepsilon.$$

This gives

$$T_{9\sigma(i)\sigma(i+1)}^3 = a'_{\sigma(i+1)} + d'_{\sigma(i)} + \max\{0, c'_{\sigma(i)} - a'_{\sigma(i+1)}, b'_{\sigma(i+1)} - d'_{\sigma(i)}\}.$$

Note that the distance between cities (parts) i and $i + 1$ depends only on the parameters of those cities. Thus, we have a TSP formulation.

Lemma 8. The part sequencing problem associated with cycle $S_{9,4}$ is unary NP-hard.

Proof. By reduction from the classical scheduling problem $F3|no-wait|C_t$. Röck [33] shows that the cycle time minimization problem in this problem is unary NP-hard. For more details, the reader may refer to the paper on no-wait scheduling by Hall and Sriskandarajah [17]. Let x_i , y_i and z_i denote the processing time of job i on the first, second and third machine, respectively, in the three-machine no-wait flowshop. In the three-machine no-wait flowshop, the cycle time, $C_{(\sigma)}$, for the production of n jobs in sequence σ can be written as follows [33]: $C_{(\sigma)} = \sum_{i=1}^n C_{\sigma(i)\sigma(i+1)}$, where $C_{\sigma(i)\sigma(i+1)} = \max\{x_{\sigma(i)} + y_{\sigma(i)} - x_{\sigma(i+1)}, x_{\sigma(i)} + y_{\sigma(i)} + z_{\sigma(i)} - x_{\sigma(i+1)} - y_{\sigma(i+1)}, x_{\sigma(i)}\}$. Rewriting, we have $C_{\sigma(i)\sigma(i+1)} = x_{\sigma(i)} + y_{\sigma(i)} - x_{\sigma(i+1)} + \max\{0, z_{\sigma(i)} - y_{\sigma(i+1)}, x_{\sigma(i+1)} - y_{\sigma(i)}\}$. Now, if we set $a'_i = y_i$, $b'_i = x_i$, $c'_i = z_i$ and $d'_i = y_i$, it is clear that minimizing $T_{9(\sigma)}$ is equivalent to minimizing $C_{(\sigma)}$. $\square$

The cycle time for the production of n parts in sequence σ under cycle $S_{10,4}$ is derived as follows:

$$T_{10\sigma(i)\sigma(i+1)}^3 = \lambda + \theta + w_1^{i+1} + w_2^{i+1} + w_3^i + w_4^i,$$

where

$$w_1^{i+1} = \max\{0, a_{\sigma(i+1)} - w_3^i - \lambda\},$$

$$w_2^{i+1} = \max\{0, b_{\sigma(i+1)} - w_4^i - \lambda\},$$

$$w_3^i = \max\{0, c_{\sigma(i)} - \lambda\},$$

$$w_4^i = \max\{0, d_{\sigma(i)} - w_1^{i+1} - \lambda\},$$

$$\lambda = 6\delta + 2\varepsilon \quad \text{and} \quad \theta = 12\delta + 8\varepsilon.$$

Therefore,

$$T_{10(\sigma)} = \sum_{i=1}^n T_{10\sigma(i)\sigma(i+1)}^3 = n(\lambda + \theta) + \sum_{i=1}^n (w_1^{i+1} + w_2^{i+1} + w_3^i + w_4^i).$$

Also,

$$w_2^{i+1} = \max\{0, b_{\sigma(i+1)} - w_4^i - \lambda\},$$

$$w_2^{i+1} + w_4^i = \max\{0, b_{\sigma(i+1)} - \lambda, d_{\sigma(i)} - w_1^{i+1} - \lambda\},$$

$$\begin{aligned}
& w_2^{i+1} + w_4^i + w_1^{i+1} \\
& = \max\{0, a_{\sigma(i+1)} - w_3^i - \lambda, d_{\sigma(i)} - \lambda, b_{\sigma(i+1)} - \lambda, a_{\sigma(i+1)} + b_{\sigma(i+1)} - w_3^i - 2\lambda\}, \\
& \text{and} \\
& w_2^{i+1} + w_4^i + w_1^{i+1} + w_3^i \\
& = \max\{0, c_{\sigma(i)} - \lambda, b_{\sigma(i+1)} - \lambda, b_{\sigma(i+1)} + c_{\sigma(i)} - 2\lambda, a_{\sigma(i+1)} - \lambda, \\
& \quad a_{\sigma(i+1)} + b_{\sigma(i+1)} - 2\lambda, d_{\sigma(i)} - \lambda, c_{\sigma(i)} + d_{\sigma(i)} - 2\lambda\}.
\end{aligned}$$

Note that the distance between cities (parts) i and $i + 1$ depends only on the parameters of those cities. Thus, we have a TSP formulation

Lemma 9. The part sequencing problem associated with cycle $S_{10,4}$ is unary NP-hard.

Proof. The proof is by reduction from problem $F3|no-wait|C_t$, and similar to that of lemma 8. By setting $a_i = x_i + \lambda$, $b_i = y_i + \lambda$, $c_i = y_i + \lambda$ and $d_i = z_i + \lambda$, we have $T_{10\sigma(i)\sigma(i+1)}^3 = \lambda + \theta + y_{\sigma(i)} + y_{\sigma(i+1)} + \max\{0, x_{\sigma(i+1)} - y_{\sigma(i)}, z_{\sigma(i)} - y_{\sigma(i+1)}\}$. Thus, minimizing $T_{10(\sigma)}$ is equivalent to minimizing $C_{(\sigma)}$. $\square$

The following two results relate the complexity of part sequencing problems in cells with m machines to that in larger cells. We consider two cases. The first case requires that $\mathcal{N} < m + 1$ in the larger cell, and is studied in lemma 10. The second case requires that $\mathcal{N} = m + 2$ in the larger cell, and is studied in lemma 11.

Lemma 10. If the part sequencing problems associated with all V2-cycles in an m machine cell, for $m \geq 4$, are unary NP-hard, then the part sequencing problems associated with all V2-cycles which have $\mathcal{N} < m + 1$ in an $m + 1$ machine cell are unary NP-hard.

Proof. Consider an arbitrary V2-cycle, $S_{g,m+1}$, with $\mathcal{N} < m + 1$ waiting times in an $m + 1$ machine cell, where $m \geq 4$. Let $M_{j_1}, \dots, M_{j_{\mathcal{N}}}$ denote the machines at which $S_{g,m+1}$ has waiting times. Note that, if $\mathcal{N} = 2$, then $2 \leq j_1, j_2 \leq m$, since otherwise $S_{g,m+1}$ is a V1-cycle. Following the discussion in section 3, let $S_{g,m+1} : \{M_{m+1}^+, \dots, M_p^-, M_q^-, M_r^-, \dots, M_{m+1}^+\}$.

Since $\mathcal{N} < m + 1$, there exists a machine M_q in $S_{g,m+1}$, where $q \in \{1, \dots, m + 1\} \setminus \{j_1, \dots, j_{\mathcal{N}}\}$, without a waiting time. If $\mathcal{N} = 2$, then since $m + 1 \geq 5$, there exists a machine M_q satisfying the above condition and also $q \neq 1$ and $q \neq m + 1$. Since the robot stays with the part between loading it on M_q and unloading it, we must have $r = q + 1$. Thus, M_{r-2}^- will occur between M_{m+1}^+ and M_p^- in $S_{g,m+1}$. We remove M_q^- from $S_{g,m+1}$, to give $S_{g,m} : \{M_{m+1}^+, \dots, M_p^-, M_r^-, \dots, M_{m+1}^+\}$.

Now consider an m machine cell containing machines $M_1, \dots, M_{q-1}, M_{q+1}, \dots, M_{m+1}$, in which $M_{q+1}, \dots, M_{m+1}$ are reindexed as $M_q, \dots, M_m$, respectively. Note that, as

a result of this reindexing, if $\mathcal{N} = 2$, then $2 \leq j_1, j_2 \leq m - 1$. Let $S_{h,m}$ denote a cycle in that cell in which the machines are loaded in the same sequence as in $S_{g,m}$. From Sethi et al. [34], that cycle is unique. First, we show that $S_{h,m}$ is a feasible V2-cycle. Let M_s denote a machine at which an E_0 state occurs in $S_{g,m+1}$. Since there is no waiting time at M_q , $s \neq q$. Since M_r is empty and M_{r-2} is occupied by a part (where we let $M_0 = I$) when activity M_p^- is performed in $S_{g,m+1}$, activity M_r^- can be performed immediately after M_p^- in $S_{g,m}$. Thus, $S_{g,m}$ is a feasible cycle in which an E_0 state occurs at M_s . Since $S_{h,m}$ has the same sequence of machine loadings as $S_{g,m}$, it is also feasible and reaches the state E_0 at the machine corresponding to machine M_s . Note that $S_{h,m}$ and $S_{g,m+1}$ have the same number of waiting times. Since $S_{h,m}$ has an E_0 state, $\mathcal{N} \geq 2$, and $2 \leq j_1, j_2 \leq m - 1$ if $\mathcal{N} = 2$, $S_{h,m}$ is a V2-cycle.

Now, using the lemma condition, we can prove unary NP-hardness of the part sequencing problem associated with cycle $S_{g,m+1}$ by reduction from that associated with cycle $S_{h,m}$. In any cycle for the $m + 1$ machine cell, the cycle time, C , can be written as $C = Z + W$, where Z is a constant and W is the total robot waiting time. Since Z is a constant, minimizing C is equivalent to minimizing W . Given an arbitrary instance of the part sequencing problem associated with cycle $S_{h,m}$, using data $p'_{i,j}$, $i = 1, \dots, n$, $j = 1, \dots, m$, δ' , ε' , and total robot waiting time W' , we construct an instance of the part sequencing problem associated with cycle $S_{g,m+1}$ as follows:

$$\begin{aligned} p_{i,j} &= p'_{i,j} + \Delta_j, & i = 1, \dots, n, \quad j = 1, \dots, q - 1, \\ p_{i,q} &= 0, & i = 1, \dots, n, \\ p_{i,j} &= p'_{i,j-1} + \Delta_j, & i = 1, \dots, n, \quad j = q + 1, \dots, m + 1, \\ \delta &= \delta', \\ \varepsilon &= \varepsilon', \\ W &= W', \end{aligned}$$

where Δ_j denotes the difference between the total travel, loading and unloading time from the loading of a part on M_j until the robot returns to M_j in the $m + 1$ machine cell, and the equivalent total time in the m machine cell. Note that Δ_j can be found in $O(m)$ time for $j = 1, \dots, m$, therefore the above construction requires polynomial time.

We show that the robot waiting time at a machine (except M_q) for a part $P_{\sigma(i)}$ in $S_{g,m+1}$ is equal to the waiting time at the corresponding machine for that part in $S_{h,m}$. Let the waiting time for part $P_{\sigma(i)}$ at M_j in $S_{g,m+1}$ be $w_j^i = \max\{0, p_{\sigma(i),j} - t_j^i - W_j^i\}$, where t_j^i is the total robot activity (travel, loading and unloading) time from the loading of part $P_{\sigma(i)}$ on M_j until the robot returns to M_j , and W_j^i is the total robot waiting time on all the machines during this interval. The waiting time w_j^i on the corresponding machine $M_{j'}$ in $S_{h,m}$ is given by $w_{j'}^i = \max\{0, p'_{\sigma(i),j'} - t_{j'}^i - W_{j'}^i\}$, where $t_{j'}^i$ and $W_{j'}^i$ are defined analogously with t_j^i and W_j^i . Since $S_{h,m}$ and $S_{g,m+1}$ have the same sequence of machine loadings, and from the definition of q , the waiting times for all the parts are zero at M_q in $S_{g,m+1}$. Thus, the number of robot waiting times

in $W_{j'}^i$ is the same as in W_j^i . Now, since $t_j^i = t_{j'}^i + \Delta_j$ from the definition of Δ_j , and since $p_{\sigma(i),j} = p'_{\sigma(i),j'} + \Delta_j$ by construction, we have $W_{j'}^i = W_j^i$ and $w_j^i = w_{j'}^i$. Thus, the part sequencing problems associated with cycles $S_{g,m+1}$ and $S_{h,m}$ are equivalent. $\square$

Lemma 11. If the part sequencing problems associated with all V2-cycles which have $\mathcal{N} = m$ in an m machine cell, for $m \geq 4$, are unary NP-hard, then the part sequencing problems associated with all V2-cycles which have $\mathcal{N} = m + 2$ in an $m + 2$ machine cell are unary NP-hard.

Proof. From the proof of lemma 5, $\mathcal{N}$ is even in all V2-cycles. Therefore, we may assume that m is even. From lemma 6, there is a unique V2-cycle in which $\mathcal{N} = m$ in an m machine cell, where m is even. Using data $p'_{i,j}$, $i = 1, \dots, n$, $j = 1, \dots, m$, δ' , ε' , and C' , its cycle time, $T_{h(\sigma)}$, can be developed as follows:

$$S_{h,m} : \{M_{m/2+1}^-, M_1^-, M_{m/2+2}^-, M_2^-, \dots, M_{m/2+j}^-, M_j^-, \dots, M_m^-, M_{m/2}^-, M_m^+, M_{m/2+1}^-\}.$$

This gives

$$T_{h(\sigma(i)\sigma(i+1))}^{m/2+1} = \frac{(m+2)^2}{2} \delta' + (2m+2) \varepsilon' + w_1^{i+1} + \dots + w_{m/2}^{i+1} + w_{m/2+1}^i + \dots + w_m^i,$$

where

$$\begin{aligned} w_j^{i+1} &= \max\{0, p'_{\sigma(i+1),j} - w_{m/2+j}^i - (m+2)\delta' - 2\varepsilon'\}, & j = 1, \dots, m/2, \\ w_{m/2+1}^i &= \max\{0, p'_{\sigma(i),m/2+1} - (m+2)\delta' - 2\varepsilon'\}, \\ w_{m/2+j+1}^i &= \max\{0, p'_{\sigma(i),m/2+j+1} - w_j^{i+1} - (m+2)\delta' - 2\varepsilon'\}, & j = 1, \dots, m/2 - 1. \end{aligned}$$

Therefore

$$\begin{aligned} T_{h(\sigma)}^{m/2+1} &= \sum_{i=1}^n T_{h(\sigma(i)\sigma(i+1))}^{m/2+1} = n \left[\frac{(m+2)^2}{2} \delta' + (2m+2) \varepsilon' \right] \\ &\quad + \sum_{i=1}^n (w_1^{i+1} + \dots + w_{m/2}^{i+1} + w_{m/2+1}^i + \dots + w_m^i). \end{aligned}$$

Given an arbitrary instance of the above part sequencing problem in an m machine cell, we construct an instance of the equivalent problem in an $m + 2$ machine cell, as follows:

$$\begin{aligned} p_{i,1} &= 0, & i = 1, \dots, n, \\ p_{i,j+1} &= p'_{i,j}, & i = 1, \dots, n, \quad j = 1, \dots, m/2, \\ p_{i,m/2+2} &= 0, & i = 1, \dots, n, \end{aligned}$$

$$\begin{aligned}
p_{i,m/2+j+2} &= p'_{i,j}, & i = 1, \dots, n, \quad j = 1, \dots, m/2, \\
\delta &= (m+2)\delta'/(m+4), \\
\varepsilon &= \varepsilon', \\
C &= C' + (m+2)\delta' + 4\varepsilon'.
\end{aligned}$$

The cycle time formula for the unique V2-cycle with $\mathcal{N} = m+2$ in an $m+2$ machine cell and $m+2$ even can be developed as follows:

$$T_{i\sigma(i)\sigma(i+1)}^{m/2+2} = \frac{(m+4)^2}{2} \delta + (2m+6)\varepsilon + w_1^{i+1} + \dots + w_{m/2+1}^{i+1} + w_{m/2+2}^i + \dots + w_{m+2}^i,$$

where

$$\begin{aligned}
w_j^{i+1} &= \max\{0, p_{\sigma(i+1),j} - w_{m/2+j+1}^i - (m+4)\delta - 2\varepsilon\}, & j = 1, \dots, m/2+1, \\
w_{m/2+2}^i &= \max\{0, p_{\sigma(i),m/2+2} - (m+4)\delta - 2\varepsilon\}, \\
w_{m/2+j+2}^i &= \max\{0, p_{\sigma(i),m/2+j+2} - w_j^{i+1} - (m+4)\delta - 2\varepsilon\}, & j = 1, \dots, m/2.
\end{aligned}$$

It is now easy to verify that the part sequencing problem associated with the unique V2-cycle in which $\mathcal{N} = m$ in the m machine cell is equivalent to that associated with the unique V2-cycle in which $\mathcal{N} = m+2$ in the $m+2$ machine cell. $\square$

4.4. Class W: NP-hard non-TSP problems

We show that if a robot move cycle satisfies neither condition C1 nor condition C3, in which case we call it a W -cycle, then the part sequencing problem associated with that cycle is unary NP-hard. In this kind of cycle, the robot frequently leaves a part to be processed at a machine and performs other activities in the cell. This creates many waiting times and, more importantly, those waiting times are recursive.

It is interesting to note that there is no analog of lemma 6 for W -cycles. That is, the same selection of $\mathcal{N}$ waiting times can appear in more than one W -cycle. For example, the W -cycles $S_{2,3}$ and $S_{6,3}$, as described in section 3, both have waiting times on all the machines in a three machine cell.

Hall et al. [14] show that the part sequencing problems associated with cycles $S_{2,3}$ and $S_{6,3}$ are unary NP-hard. The following two results relate the complexity of part sequencing problems in cells with m machines to that in larger cells. We consider two cases. The first case requires that $\mathcal{N} < m+1$ in the larger cell, and is studied in lemma 12. The second case requires that $\mathcal{N} = m+1$ in the larger cell, and is studied in lemma 13.

Lemma 12. If the part sequencing problems associated with all W -cycles in an m machine cell, for $m \geq 3$, are unary NP-hard, then the part sequencing problems associated with all W -cycles which have $\mathcal{N} < m+1$ in an $m+1$ machine cell are unary NP-hard.

Proof. First, consider the case $\mathcal{N} = 1$. Here the robot loads a part, and leaves it at exactly one machine, M_q . Since the robot remains with the part at all other machines, it can never return and pick up the part at M_q . Consequently, there are no robot move cycles in which $\mathcal{N} = 1$. The contrapositive of corollary 1 implies that there is no W -cycle in which $\mathcal{N} = 2$. We therefore consider an arbitrary W -cycle, $S_{g,m+1}$, with $3 \leq \mathcal{N} < m + 1$ waiting times in an $m + 1$ machine cell, where $m \geq 3$. Let $M_{j_1}, \dots, M_{j_{\mathcal{N}}}$ denote the machines at which $S_{g,m+1}$ has waiting times. Let $S_{g,m+1} : \{M_{m+1}^+, \dots, M_p^-, M_q^-, M_r^-, \dots, M_{m+1}^+\}$. Since $\mathcal{N} < m + 1$, there exists a machine M_q in $S_{g,m+1}$, where $q \in \{1, \dots, m + 1\} \setminus \{j_1, \dots, j_{\mathcal{N}}\}$, without a waiting time. Since the robot stays with the part between loading it on M_q and unloading it, we must have $r = q + 1$. Thus, M_{r-2}^- will occur between M_{m+1}^+ and M_p^- in $S_{g,m+1}$. We remove M_q^- from $S_{g,m+1}$, to give cycle $S_{g,m} : \{M_{m+1}^+, \dots, M_p^-, M_r^-, \dots, M_{m+1}^+\}$.

Now consider an m machine cell containing machines $M_1, \dots, M_{q-1}, M_{q+1}, \dots, M_{m+1}$, in which $M_{q+1}, \dots, M_{m+1}$ are reindexed as $M_q, \dots, M_m$, respectively. Let $S_{h,m}$ denote the unique cycle in that cell in which the machines are loaded in the same sequence as in $S_{g,m}$. First, we show that $S_{h,m}$ is a feasible W -cycle. Since $S_{g,m+1}$ is a W -cycle, it does not contain an E_0 state. Since M_r is empty and M_{r-2} is occupied by a part (where we let $M_0 = I$) when activity M_p^- is performed in $S_{g,m+1}$, M_r^- can be performed immediately after M_p^- in $S_{g,m}$. Thus, $S_{g,m}$ is a feasible cycle which does not contain an E_0 state, since the sequence of machine loadings on the available machines is the same as in $S_{g,m+1}$. Therefore, at each machine loading in $S_{g,m}$, there is at least one other machine which is occupied by a part. Since $S_{h,m}$ has the same sequence of machine loadings as $S_{g,m}$, it is also feasible and does not contain an E_0 state. Note that $S_{h,m}$ and $S_{g,m+1}$ have the same number of waiting times. Since $S_{h,m}$ does not contain an E_0 state, $S_{h,m}$ is a W -cycle.

The remainder of the proof is similar to that of lemma 10. $\square$

Lemma 13. If the part sequencing problems associated with all W -cycles which have $\mathcal{N} = m$ in an m machine cell, for $m \geq 3$, are unary NP-hard, then the part sequencing problems associated with all W -cycles which have $\mathcal{N} = m + 1$ in an $m + 1$ machine cell are unary NP-hard.

Proof. Consider an arbitrary W -cycle, $S_{g,m+1}$, in an $m + 1$ machine cell. We first identify into which of the following two types of cycle $S_{g,m+1}$ falls:

Type A: cycle $S_{g,m+1}$ contains the sequence of machine loadings M_i^-, M_1^-, M_{i+1}^- for some $2 \leq i \leq m + 1$, where we let $M_{m+2}^- = M_{m+1}^+$.

Type B: otherwise.

We first consider an arbitrary Type B cycle, $S_{g,m+1} : \{M_{m+1}^+, \dots, M_p^-, M_1^-, M_r^-, \dots, M_{m+1}^+\}$, where $r \neq p + 1$. We let $q = 1$ and remove M_1^- from cycle $S_{g,m+1}$, thus obtaining the cycle $S_{g,m} : \{M_{m+1}^+, \dots, M_p^-, M_r^-, \dots, M_{m+1}^+\}$. Now consider an m machine cell

containing machines $M_2, \dots, M_{m+1}$, in which those machines are reindexed as $M_1, \dots, M_m$, respectively. Let $S_{h,m}$ denote the unique cycle in that cell in which the machines are loaded in the same sequence as in $S_{g,m}$ and $\mathcal{N} = m$. Thus, $S_{h,m} : \{M_m^+, \dots, M_{p-1}^-, M_{r-1}^-, \dots, M_m^+\}$. Since removing M_1^- from $S_{g,m+1}$ will not create a cycle of the form $\{\dots, M_i^-, M_{i+1}^-, \dots\}$ (where we let $M_{m+1}^- = M_m^+$), for some $1 \leq i \leq m$, it follows that $\mathcal{N} = m$ in $S_{h,m}$. Since machine M_{r-1} is occupied by a part and machine M_r is free when machine M_p is loaded in $S_{g,m+1}$, activity M_r^- can be performed immediately after M_p^- in $S_{g,m}$. Thus, $S_{g,m}$ is a feasible cycle. Since $S_{h,m}$ has the same sequence of machine loadings as $S_{g,m}$, it follows that $S_{h,m}$ is also a feasible cycle. As in the proof of lemma 12, $S_{h,m}$ does not contain an E_0 state, and is thus a W -cycle with $\mathcal{N} = m$.

Consider an arbitrary cycle of Type A in an m' machine cell, where $m' \geq 4$. We consider two subtypes, which we denote by A1 and A2.

Type A1: $m' = 4$.

Here we let $m + 1 = 4$, and provide a transformation to a cell with $m = 3$ machines. From appendix A, when $m + 1 = 4$, there are only two W -cycles:

$$S_{20,4} : \{M_4^+, M_4^-, M_2^-, M_1^-, M_3^-, M_4^+\},$$

$$S_{22,4} : \{M_4^+, M_3^-, M_2^-, M_4^-, M_1^-, M_4^+\}.$$

Let $S_{g,m+1} : \{M_{m+1}^+, \dots, M_p^-, M_i^-, M_1^-, M_{i+1}^-, M_r^-, \dots, M_{m+1}^+\}$ be a W -cycle in this class. The following transformation obtains a feasible cycle with $\mathcal{N} = m$ in the m machine cell. Let $q = i$. Thus, $q = 2$ in $S_{20,4}$, and $q = 4$ in $S_{22,4}$. We remove M_q^- from cycle $S_{g,m+1}$ and interchange the positions of M_{q+1}^- and M_1^- , (where we let $M_{m+2}^- = M_{m+1}^+$ in $S_{22,4}$). Then the following cycle is obtained: $S_{g,m} : \{M_{m+1}^+, \dots, M_p^-, M_{i+1}^-, M_1^-, M_r^-, \dots, M_{m+1}^+\}$. Now consider an m machine cell containing machines $M_1, \dots, M_{q-1}, M_{q+1}, \dots, M_{m+1}$, in which $M_{q+1}, \dots, M_{m+1}$ are reindexed as $M_q, \dots, M_m$, respectively. Let $S_{h,m}$ denote the unique cycle in that cell in which the machines are loaded in the same sequence as in $S_{g,m}$. When activity M_p^- is performed in $S_{g,m+1}$, the following conditions must be satisfied. First, machines M_i and M_{i+1} must be free. Second, machine M_{i-1} must be occupied by a part. Also, by definition of $S_{g,m+1}$, $p \neq i - 1$. Finally, note that in both $S_{20,4}$ and $S_{22,4}$, we have $r \neq 2$. Under these conditions, the machine loading sequence $\{M_p^-, M_{i+1}^-, M_1^-, M_r^-\}$ is feasible in $S_{g,m}$, and thus $S_{g,m}$ is a feasible cycle. Further, since it has the same sequence of machine loadings as $S_{g,m}$, $S_{h,m}$ is also a feasible cycle. As above, $S_{h,m}$ does not contain an E_0 state, and is thus a W -cycle with $\mathcal{N} = m$. Using the above transformation and the following construction, cycles $S_{6,3}$ and $S_{2,3}$ are reducible to cycles $S_{20,4}$ and $S_{22,4}$, respectively.

Given an arbitrary instance of the part sequencing problem of Type A1 or Type B associated with cycle $S_{h,m}$, using data $p'_{i,j}$, $i = 1, \dots, n$, $j = 1, \dots, m$, δ' , ε' , and the total robot waiting time W' , we construct an instance of the part sequencing problem associated with cycle $S_{g,m+1}$ as follows:

$$\begin{aligned}
p_{i,j} &= p'_{i,j} + \Delta_j, & i = 1, \dots, n, \quad j = 1, \dots, q-1, \\
p_{i,q} &= 0, & i = 1, \dots, n, \\
p_{i,j} &= p'_{i,j-1} + \Delta_j, & i = 1, \dots, n, \quad j = q+1, \dots, m+1, \\
\delta &= \delta', \\
\varepsilon &= \varepsilon', \\
W &= W',
\end{aligned}$$

where Δ_j is defined in the proof of lemma 10.

The remainder of the proof is similar to that of lemma 10.

Type A2: $m' > 4$.

Here we let $m+2 = m'$, and provide a transformation to a cell with m machines. Let $S_{g,m+2} : \{M_{m+2}^+, \dots, M_p^-, M_i^-, M_1^-, M_{i+1}^-, M_r^-, \dots, M_{m+2}^+\}$. By definition, $p \neq i-1$ and $r \neq i+2$. Here, we need the following transformation to obtain a feasible cycle with $\mathcal{N} = m$ in the m machine cell. Let $q = i$. We remove M_i^- and M_1^- from cycle $S_{g,m+2}$. Then the following cycle is obtained: $S_{g,m} : \{M_{m+2}^+, \dots, M_p^-, M_{i+1}^-, M_r^-, \dots, M_{m+2}^+\}$. Now consider an m machine cell containing machines $M_2, \dots, M_{q-1}, M_{q+1}, \dots, M_{m+2}$, in which $M_2, \dots, M_{q-1}$ are reindexed as $M_1, \dots, M_{q-2}$, respectively, and $M_{q+1}, \dots, M_{m+2}$ are reindexed as $M_{q-1}, \dots, M_m$, respectively. Let $S_{h,m}$ denote the unique cycle in that cell in which the machines are loaded in the same sequence as in $S_{g,m}$. When activity M_p^- is performed in $S_{g,m+2}$, the following conditions must be satisfied. First, machines M_i and M_{i+1} must be free. Second, machine M_{i-1} must be occupied by a part. Under these conditions, activity M_{i+1}^- can be performed immediately after M_p^- in $S_{g,m}$, thus $S_{g,m}$ is a feasible cycle. Further, since it has the same sequence of machine loadings as $S_{g,m}$, $S_{h,m}$ is also a feasible cycle. As above, $S_{h,m}$ does not contain an E_0 state, and is thus a W -cycle with $\mathcal{N} = m$.

Given an arbitrary instance of the part sequencing problem of Type A2 associated with cycle $S_{h,m}$, using data $p'_{i,j}$, $i = 1, \dots, n$, $j = 1, \dots, m$, δ' , ε' , and the total robot waiting time W' , we construct an instance of the part sequencing problem associated with cycle $S_{g,m+2}$ as follows:

$$\begin{aligned}
p_{i,1} &= 0, & i = 1, \dots, n, \\
p_{i,j} &= p'_{i,j} + \Delta_j, & i = 1, \dots, n, \quad j = 2, \dots, q-1, \\
p_{i,q} &= 0, & i = 1, \dots, n, \\
p_{i,j} &= p'_{i,j-1} + \Delta_j, & i = 1, \dots, n, \quad j = q+1, \dots, m+2, \\
\delta &= \delta', \\
\varepsilon &= \varepsilon', \\
W &= W',
\end{aligned}$$

where Δ_j is defined in the proof of lemma 10.

The remainder of the proof is similar to that of lemma 10. □

4.5. Overview

The main result of the paper now follows.

Theorem 1. In an m machine robotic cell with $m \geq 4$, there are $m!$ robot move cycles, of which

- (a) one U -cycle defines a trivially solvable part sequencing problem;
- (b) $(2m - 3)$ $V1$ -cycles define a part sequencing problem which is solvable in $O(n \log n)$ time;

(c)
$$\sum_{t=1}^{\lfloor m/2 \rfloor} \binom{m}{2t} - 2m + 3$$

$V2$ -cycles define a part sequencing problem which can be formulated as a TSP, and which is unary NP-hard.

(d)
$$m! - 1 - \sum_{t=1}^{\lfloor m/2 \rfloor} \binom{m}{2t}$$

W -cycles define a part sequencing problem which in general cannot be formulated as a TSP, and which is unary NP-hard.

Proof. Sethi et al. [34] show that there are $m!$ potentially optimal robot move cycles in an m machine cell.

- (a) Follows from lemma 2.
- (b) Follows from lemma 3.
- (c) The $(\Rightarrow)$ part of lemma 4 shows that the associated part sequencing problems can each be formulated as a TSP. The proof of NP-hardness is by induction on m . Lemmas 8 and 9 provide the basis for $m = 4$. Lemma 10 provides the induction step for $\mathcal{N} < m + 1$. Since by assumption there exists a $V2$ -cycle with $\mathcal{N} = m$, and since from the proof of lemma 5 $\mathcal{N}$ is even in all $V2$ -cycles, there are no $V2$ -cycles with $\mathcal{N} = m + 1$. Lemma 11 provides the induction step for $\mathcal{N} = m + 2$. The number of cycles follows from lemmas 3 and 7.
- (d) The lack of a TSP formulation follows from the $(\Leftarrow)$ part of lemma 4. The proof of NP-hardness is by induction on m . Hall et al. [14] provide the basis for $m = 3$. Lemma 12 provides the induction step for $\mathcal{N} < m + 1$. Lemma 13 provides the induction step for $\mathcal{N} = m + 1$. The number of cycles follows from the formula for the total number of cycles in Sethi et al. [34], and from the earlier parts of the theorem. $\square$

Recall that Sethi et al. [34] (respectively, Hall et al. [13, 14]) resolve the complexity of the part sequencing problems associated with the two (resp., six) robot move

cycles in a two (resp., three) machine cell. Therefore, those results and theorem 1 resolve the complexity of the part sequencing problems associated with all one-unit robot move cycles in any cell. It is interesting to note, as discussed in section 4.3, that there are no V2-cycles in cells with fewer than four machines. This is the reason why the basis of the induction proof uses $m = 4$ in part (c) and $m = 3$ in part (d) of theorem 1.

As an illustration of theorem 1, for the $m! = 24$ cycles in a four-machine cell (see appendix A), one U -cycle has an associated part sequencing problem which is sequence independent; seven V -cycles have associated part sequencing problems which can be formulated as a TSP (of these, five are $V1$ -cycles which are solvable by the algorithm of Gilmore and Gomory [9], and the remaining two are unary NP-hard $V2$ -cycles); and 16 W -cycles have unary NP-hard associated part sequencing problems without TSP structure.

Finally, it is interesting to note that the recognition versions of the part sequencing problems associated with the NP-hard $V2$ -and W -cycles are in the class NP, since given a part sequence, both the cycle time and the robot waiting times can be calculated using linear programming.

5. Conclusions

In this paper, we study the problem of maximizing the steady-state throughput rate of an m machine robotic cell without buffers that produces multiple-part types. The part sequencing problems associated with all the potentially optimal one-unit robot move cycles are classified into four categories of tractability. Since one-unit robot move cycles are the easiest to implement and control, our work has direct implications for the practical operation of robotic cells.

We now suggest some directions for future research. The development of analytical methods and efficient heuristics to find good solutions for a general m machine robotic cell with multiple-part types is important. Also, the consideration of additional practical complexities, such as stochastic processing times, buffers within the cell, robots with dual grippers, and multiple robots, would be useful for industry. Scheduling problems arising in other shop configurations with a robot server are also worthy of consideration. A parallel machine environment has been studied by Hall et al. [16], but job-shop and open-shop environments have not been considered.

Appendix A: Construction of robot move cycles for $m = 4$

$$\begin{aligned}
 S_{1,4} &: \{M_4^+, M_1^-, M_2^-, M_3^-, M_4^-, M_4^+\}, & S_{2,4} &: \{M_4^+, M_1^-, M_2^-, M_4^-, M_3^-, M_4^+\}, \\
 S_{3,4} &: \{M_4^+, M_1^-, M_4^-, M_2^-, M_3^-, M_4^+\}, & S_{4,4} &: \{M_4^+, M_4^-, M_1^-, M_2^-, M_3^-, M_4^+\}, \\
 S_{5,4} &: \{M_4^+, M_1^-, M_3^-, M_2^-, M_4^-, M_4^+\}, & S_{6,4} &: \{M_4^+, M_1^-, M_3^-, M_4^-, M_2^-, M_4^+\},
 \end{aligned}$$

$$\begin{aligned}
S_{7,4} &: \{M_4^+, M_1^-, M_4^-, M_3^-, M_2^-, M_4^+\}, & S_{8,4} &: \{M_4^+, M_4^-, M_1^-, M_3^-, M_2^-, M_4^+\}, \\
S_{9,4} &: \{M_4^+, M_3^-, M_1^-, M_2^-, M_4^-, M_4^+\}, & S_{10,4} &: \{M_4^+, M_3^-, M_1^-, M_4^-, M_2^-, M_4^+\}, \\
S_{11,4} &: \{M_4^+, M_3^-, M_4^-, M_1^-, M_2^-, M_4^+\}, & S_{12,4} &: \{M_4^+, M_4^-, M_3^-, M_1^-, M_2^-, M_4^+\}, \\
S_{13,4} &: \{M_4^+, M_2^-, M_3^-, M_1^-, M_4^-, M_4^+\}, & S_{14,4} &: \{M_4^+, M_2^-, M_3^-, M_4^-, M_1^-, M_4^+\}, \\
S_{15,4} &: \{M_4^+, M_2^-, M_4^-, M_3^-, M_1^-, M_4^+\}, & S_{16,4} &: \{M_4^+, M_4^-, M_2^-, M_3^-, M_1^-, M_4^+\}, \\
S_{17,4} &: \{M_4^+, M_2^-, M_1^-, M_3^-, M_4^-, M_4^+\}, & S_{18,4} &: \{M_4^+, M_2^-, M_1^-, M_4^-, M_3^-, M_4^+\}, \\
S_{19,4} &: \{M_4^+, M_2^-, M_4^-, M_1^-, M_3^-, M_4^+\}, & S_{20,4} &: \{M_4^+, M_4^-, M_2^-, M_1^-, M_3^-, M_4^+\}, \\
S_{21,4} &: \{M_4^+, M_3^-, M_2^-, M_1^-, M_4^-, M_4^+\}, & S_{22,4} &: \{M_4^+, M_3^-, M_2^-, M_4^-, M_1^-, M_4^+\}, \\
S_{23,4} &: \{M_4^+, M_3^-, M_4^-, M_2^-, M_1^-, M_4^+\}, & S_{24,4} &: \{M_4^+, M_4^-, M_3^-, M_2^-, M_1^-, M_4^+\}.
\end{aligned}$$

Appendix B: Cycle time derivations for $m = 3$

We now derive the cycle times for cycles $S_{1,3}, \dots, S_{6,3}$ in $MRC3|no-buffer, k \geq 2, 1-unit|C_t$, starting from a convenient state in each case.

Starting from the initial state $E = (\Omega, \phi, \phi, M_1^-)$, where the robot has just loaded part $P_{\sigma(i)}$ on M_1 , and machines M_2 and M_3 are free, we can derive the following expression for $S_{1,3}$:

$$T_{1\sigma(i)\sigma(i+1)}^1 = \alpha_1 + a_{\sigma(i)} + b_{\sigma(i)} + c_{\sigma(i)},$$

where $\alpha_1 = 4\delta_0 + 2\sum_{i=1}^4 \delta_i + \sum_{i=1}^8 \varepsilon_i - 3\gamma$. We obtain

$$T_{1(\sigma)} = \sum_{i=1}^n T_{1\sigma(i+1)\sigma(i+2)}^1 = n\alpha_1 + \sum_{i=1}^n (a_i + b_i + c_i).$$

Starting from the initial state $E = (\phi, \Omega, \Omega, M_2^-)$, where the robot has just loaded part $P_{\sigma(i)}$ on machine M_2 , M_3 is occupied by part $P_{\sigma(i-1)}$, and M_1 is free, we can derive the following expression for $S_{2,3}$:

$$T_{2\sigma(i)\sigma(i+1)}^2 = \sum_{i=1}^8 \varepsilon_i + 8\delta_0 + 2\delta_1 + 4\delta_2 + 4\delta_3 + 2\delta_4 - 4\gamma + w_1^{i+1} + w_2^i + w_3^{i-1},$$

where $w_1^{i+1} = \max\{0, a_{2\sigma(i+1)} - \beta_2 - w_2^i\}$, $w_2^i = \max\{0, b_{2\sigma(i)} - \beta_2 - w_3^{i-1}\}$, and $w_3^{i-1} = \max\{0, c_{2\sigma(i-1)} - \beta_2 - w_1^i\}$. Substituting for w_1^{i+1} and w_2^i , we obtain

$$T_{2(\sigma)} = \sum_{i=1}^n T_{2\sigma(i)\sigma(i+1)}^2 = n\alpha_2 + \sum_{i=1}^n \max\{\beta_2, a_{2\sigma(i+1)}, b_{2\sigma(i)} - w_3^{i-1}\} + \sum_{i=1}^n w_3^{i-1},$$

where $w_3^{i-1} = \max\{0, c_{2\sigma(i-1)} - \max\{\beta_2, a_{2\sigma(i)} - \max\{0, b_{2\sigma(i-1)} - \beta_2 - w_3^{i-2}\}\}\}$, where we define $b_{2\sigma(n+1)} = b_{2\sigma(1)}$, $a_{2\sigma(n+1)} = a_{2\sigma(1)}$, $a_{2\sigma(n+2)} = a_{2\sigma(2)}$, and where $\alpha_2 = 4\delta_0 + 2\delta_2 + 2\delta_3 - \gamma$, $\beta_2 = \sum_{i=1}^8 \varepsilon_i + 4\delta_0 + 2\delta_1 + 2\delta_2 + 2\delta_3 + 2\delta_4 - 3\gamma$, $a_{2\sigma(i)} = a_{\sigma(i)} + \sum_{i=1}^4 \varepsilon_i +$

$\sum_{i=7}^8 \varepsilon_i + 2\delta_1 + 2\delta_4 - 2\gamma$, $b_{2\sigma(i)} = b_{\sigma(i)} + \sum_{i=3}^6 \varepsilon_i$, and $c_{2\sigma(i)} = c_{\sigma(i)} + \sum_{i=1}^2 \varepsilon_i + \sum_{i=5}^8 \varepsilon_i + 2\delta_1 + 2\delta_4 - 2\gamma$.

Starting from the initial state $E = (\phi, \phi, \Omega, M_3^-)$, where the robot has just loaded part $P_{\sigma(i)}$ on machine M_3 , and M_1 and M_2 are free, we can derive the following expression for $S_{3,3}$:

$$T_{3\sigma(i)\sigma(i+1)}^3 = a_{\sigma(i+1)} + 6\delta_0 + 2\delta_1 + 2\delta_2 + 4\delta_3 + 2\delta_4 + \sum_{i=1}^8 \varepsilon_i - 3\gamma + w_2^{i+1} + w_3^i,$$

where $w_2^{i+1} = \max\{0, b_{\sigma(i+1)} - 2\delta_0 - 2\delta_3 - 2\delta_4 - \varepsilon_7 - \varepsilon_8 + \gamma - w_3^i\}$, and $w_3^i = \max\{0, c_{\sigma(i)} - a_{\sigma(i+1)} - 4\delta_0 - 2\delta_1 - 2\delta_2 - 2\delta_3 - \varepsilon_1 - \varepsilon_2 - \varepsilon_3 - \varepsilon_4 + 2\gamma\}$

$$\Rightarrow T_{3\sigma(i)\sigma(i+1)}^3 = \alpha_3 + \max\{\beta_3 + a_{\sigma(i+1)}, b_{3\sigma(i+1)} + a_{\sigma(i+1)}, c_{3\sigma(i)}\},$$

where $\alpha_3 = 2\delta_0 + 2\delta_3 + \varepsilon_5 + \varepsilon_6 - \gamma$, $\beta_3 = 4\delta_0 + 2\delta_1 + 2\delta_2 + 2\delta_3 + 2\delta_4 + \varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4 + \varepsilon_7 + \varepsilon_8 - 2\gamma$, $b_{3\sigma(i)} = b_{\sigma(i)} + 2\delta_0 + 2\delta_1 + 2\delta_2 + \varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4 - \gamma$, and $c_{3\sigma(i)} = c_{\sigma(i)} + 2\delta_4 + \varepsilon_7 + \varepsilon_8$.

Thus, we obtain the cycle time, $T_{3(\sigma)}$ as follows:

$$T_{3(\sigma)} = \sum_{i=1}^n T_{3\sigma(i)\sigma(i+1)}^3 = n\alpha_3 + \sum_{i=1}^n \max\{\beta_3 + a_{\sigma(i+1)}, b_{3\sigma(i+1)} + a_{\sigma(i+1)}, c_{3\sigma(i)}\}.$$

Starting from the initial state $E = (\phi, \phi, \Omega, M_3^-)$, where the robot has just loaded part $P_{\sigma(i)}$ on machine M_3 , and M_1 and M_2 are free, we can derive the following expression for $S_{4,3}$:

$$T_{4\sigma(i)\sigma(i+1)}^3 = b_{\sigma(i+1)} + 8\delta_0 + 2\delta_1 + 4\delta_2 + 4\delta_3 + 2\delta_4 + \sum_{i=1}^8 \varepsilon_i - 5\gamma + w_1^{i+1} + w_3^i,$$

where $w_1^{i+1} = \max\{0, a_{\sigma(i+1)} - w_3^i - 4\delta_0 - 2\delta_2 - 2\delta_3 - 2\delta_4 - \varepsilon_7 - \varepsilon_8 + 3\gamma\}$, and $w_3^i = \max\{0, c_{\sigma(i)} - 4\delta_0 - 2\delta_1 - 2\delta_2 - 2\delta_3 - \varepsilon_1 - \varepsilon_2 + 3\gamma\}$. This gives

$$T_{4\sigma(i)\sigma(i+1)}^3 = \alpha_4 + b_{\sigma(i+1)} + \max\{\beta_4, c_{4\sigma(i)}, a_{4\sigma(i+1)}\},$$

where $\alpha_4 = 4\delta_0 + 2\delta_2 + 2\delta_3 + \varepsilon_3 + \varepsilon_4 + \varepsilon_5 + \varepsilon_6 - 2\gamma$, $\beta_4 = 4\delta_0 + 2\delta_1 + 2\delta_2 + 2\delta_3 + 2\delta_4 + \varepsilon_1 + \varepsilon_2 + \varepsilon_7 + \varepsilon_8 - 3\gamma$, $a_{4\sigma(i)} = a_{\sigma(i)} + 2\delta_1 + \varepsilon_1 + \varepsilon_2$, and $c_{4\sigma(i)} = c_{\sigma(i)} + 2\delta_4 + \varepsilon_7 + \varepsilon_8$.

Thus, we obtain the cycle time, $T_{4(\sigma)}$ as follows:

$$T_{4(\sigma)} = \sum_{i=1}^n T_{4\sigma(i)\sigma(i+1)}^3 = n\alpha_4 + \sum_{i=1}^n b_i + \sum_{i=1}^n \max\{\beta_4, c_{4\sigma(i)}, a_{4\sigma(i+1)}\}.$$

Starting from the initial state $E = (\phi, \Omega, \phi, M_2^-)$, where the robot has just loaded part $P_{\sigma(i)}$ on machine M_2 , and M_1 and M_3 are free, we can derive the following expression for $S_{5,3}$:

$$T_{5\sigma(i)\sigma(i+1)}^2 = c_{\sigma(i)} + 6\delta_0 + 2\delta_1 + 4\delta_2 + 2\delta_3 + 2\delta_4 + \sum_{i=1}^8 \varepsilon_i - 3\gamma + w_1^{i+1} + w_2^i,$$

where $w_1^{i+1} = \max\{0, a_{\sigma(i+1)} - w_2^i - 4\delta_0 - 2\delta_2 - 2\delta_3 - 2\delta_4 - \varepsilon_5 - \varepsilon_6 - \varepsilon_7 - \varepsilon_8 + 2\gamma - c_{\sigma(i)}\}$, and $w_2^i = \max\{0, b_{\sigma(i)} - 2\delta_0 - 2\delta_1 - 2\delta_2 - \varepsilon_1 - \varepsilon_2 + \gamma\}$. This gives

$$T_{5\sigma(i)\sigma(i+1)}^2 = \alpha_5 + \max\{\beta_5 + c_{\sigma(i)}, b_{5\sigma(i)} + c_{\sigma(i)}, a_{5\sigma(i+1)}\},$$

where $\alpha_5 = 2\delta_0 + 2\delta_2 + \varepsilon_3 + \varepsilon_4 - \gamma$, $\beta_5 = 4\delta_0 + 2\delta_1 + 2\delta_2 + 2\delta_3 + 2\delta_4 + \varepsilon_1 + \varepsilon_2 + \varepsilon_5 + \varepsilon_6 + \varepsilon_7 + \varepsilon_8 - 2\gamma$, $a_{5\sigma(i)} = a_{\sigma(i)} + 2\delta_1 + \varepsilon_1 + \varepsilon_2$, and $b_{5\sigma(i)} = b_{\sigma(i)} + 2\delta_0 + 2\delta_3 + 2\delta_4 + \varepsilon_5 + \varepsilon_6 + \varepsilon_7 + \varepsilon_8 - \gamma$.

Thus, we obtain the cycle time, $T_{5(\sigma)}$ as follows:

$$T_{5(\sigma)} = \sum_{i=1}^n T_{5\sigma(i)\sigma(i+1)}^2 = n\alpha_5 + \sum_{i=1}^n \max\{\beta_5 + c_{\sigma(i)}, b_{5\sigma(i)} + c_{\sigma(i)}, a_{5\sigma(i+1)}\}.$$

Starting from the initial state $E = (\Omega, \Omega, \Omega, M_1^-)$, where the robot has just loaded part $P_{\sigma(i)}$ on machine M_1 , and M_2 and M_3 are occupied by parts $P_{\sigma(i-1)}$ and $P_{\sigma(i-2)}$, respectively, we can derive the following expression for $S_{6,3}$:

$$T_{6\sigma(i)\sigma(i+1)}^1 = \sum_{i=1}^8 \varepsilon_i + 8\delta_0 + 2\delta_1 + 4\delta_2 + 4\delta_3 + 2\delta_4 - 4\gamma + w_1^i + w_2^{i-1} + w_3^{i-2},$$

where $w_1^i = \max\{0, a_{6\sigma(i)} - \beta_6 - w_2^{i-1} - w_3^{i-2}\}$, $w_2^{i-1} = \max\{0, b_{6\sigma(i-1)} - \beta_6 - w_3^{i-2}\}$, and $w_3^{i-2} = \max\{0, c_{6\sigma(i-2)} - \beta_6 - w_1^{i-1}\}$. Substituting for w_1^i, w_2^{i-1} and w_3^{i-2} , we obtain

$$T_{6(\sigma)} = \sum_{i=1}^n T_{6\sigma(i)\sigma(i+1)}^1 = n\alpha_6 + \sum_{i=1}^n \max\{\beta_6, c_{6\sigma(i-2)} - w_1^{i-1}, b_{6\sigma(i-1)}, a_{6\sigma(i)}\},$$

and

$$w_1^{i-1} = \max\{0, a_{6\sigma(i-1)} - \max\{\beta_6, b_{6\sigma(i-2)}, c_{6\sigma(i-3)} - w_1^{i-2}\}\},$$

where $\alpha_6 = 2\delta_0$, $\beta_6 = 6\delta_0 + 2\delta_1 + 4\delta_2 + 4\delta_3 + 2\delta_4 + \sum_{i=1}^8 \varepsilon_i - 4\gamma$, $a_{6\sigma(i)} = a_{\sigma(i)} + 2\delta_1 + 2\delta_2 + \sum_{i=1}^4 \varepsilon_i - \gamma$, $b_{6\sigma(i)} = b_{\sigma(i)} + 2\delta_0 + 2\delta_2 + 2\delta_3 + \sum_{i=3}^6 \varepsilon_i - \gamma$, and $c_{6\sigma(i)} = c_{\sigma(i)} + 2\delta_3 + 2\delta_4 + \sum_{i=5}^8 \varepsilon_i - \gamma$.

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